Efficient and Provably Safe Autonomous Intersection Coordination

Combinatorial Learning for Vehicle Motion Planning

Jeroen van Riel

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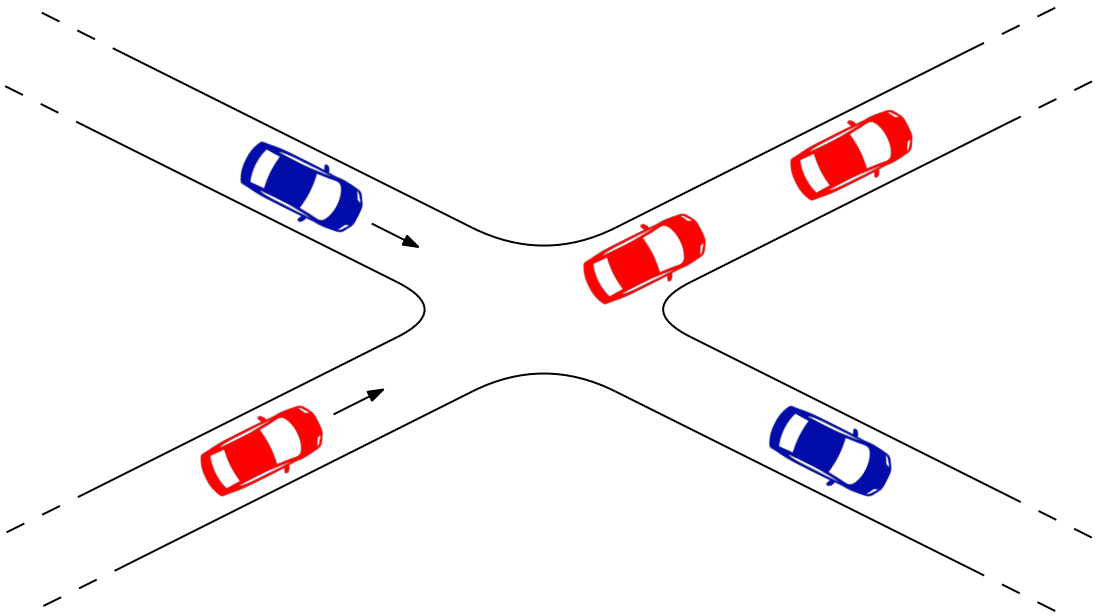
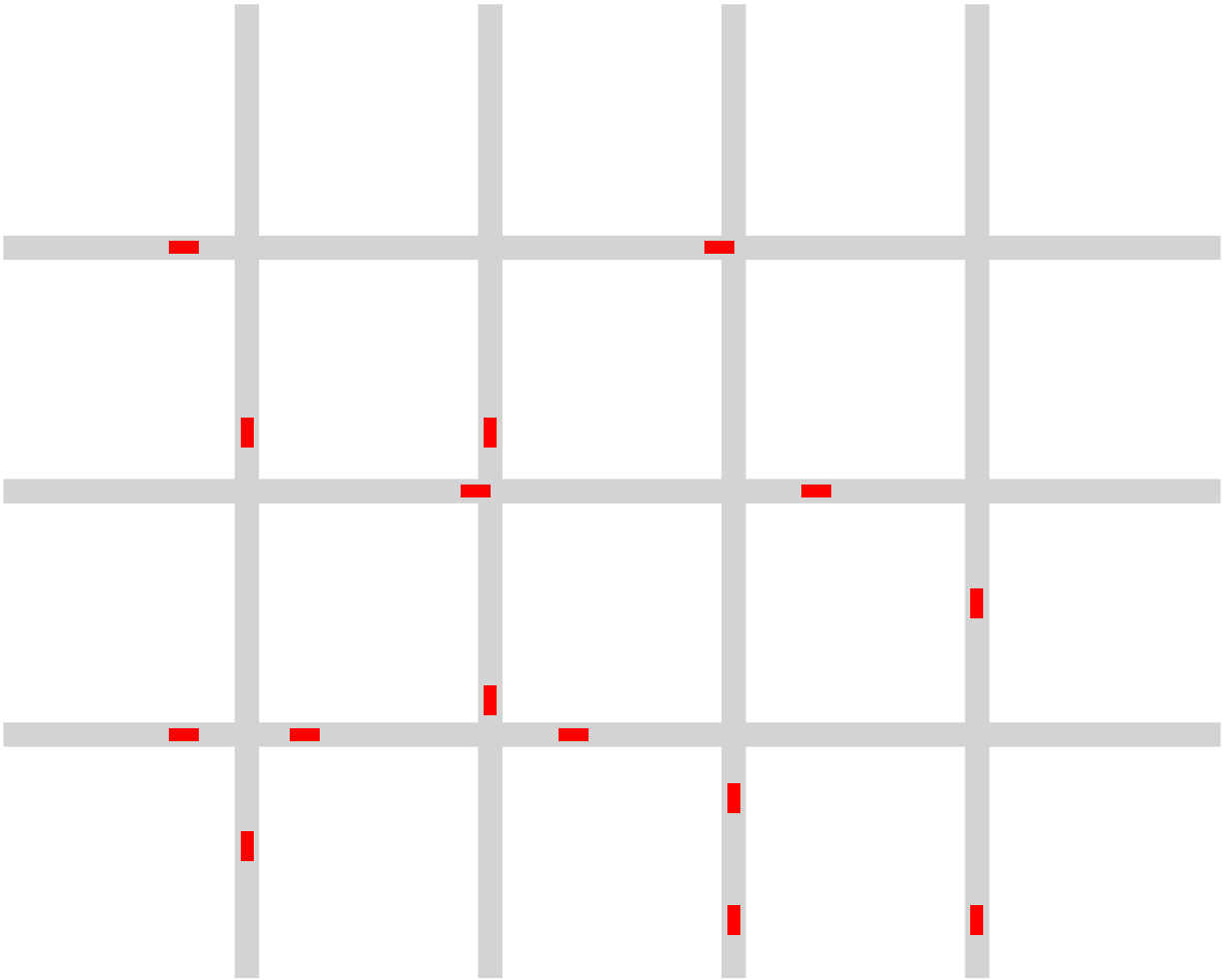
Motivation



+ safety
+ comfort

− travel time
− fuel usage

Network Model



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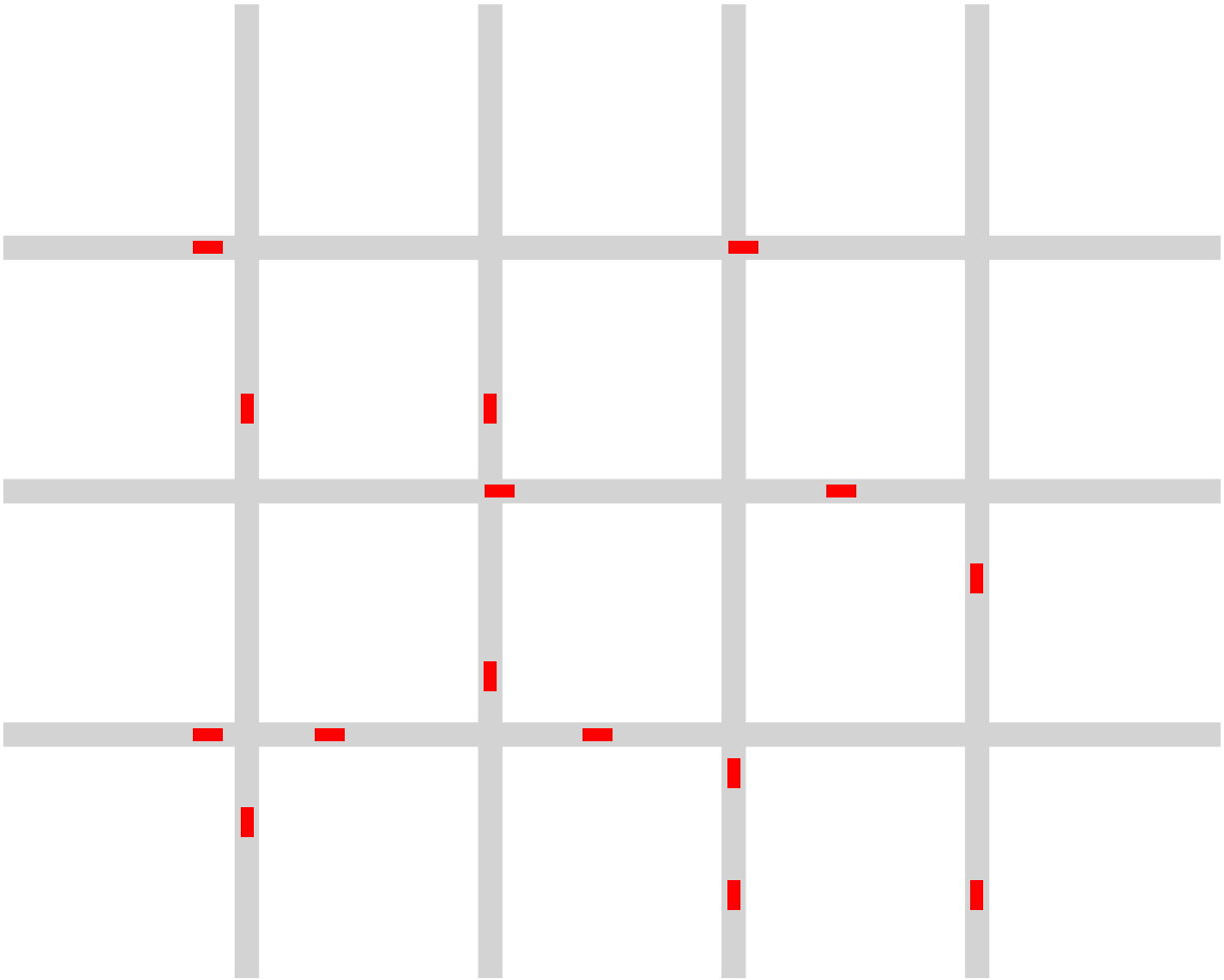

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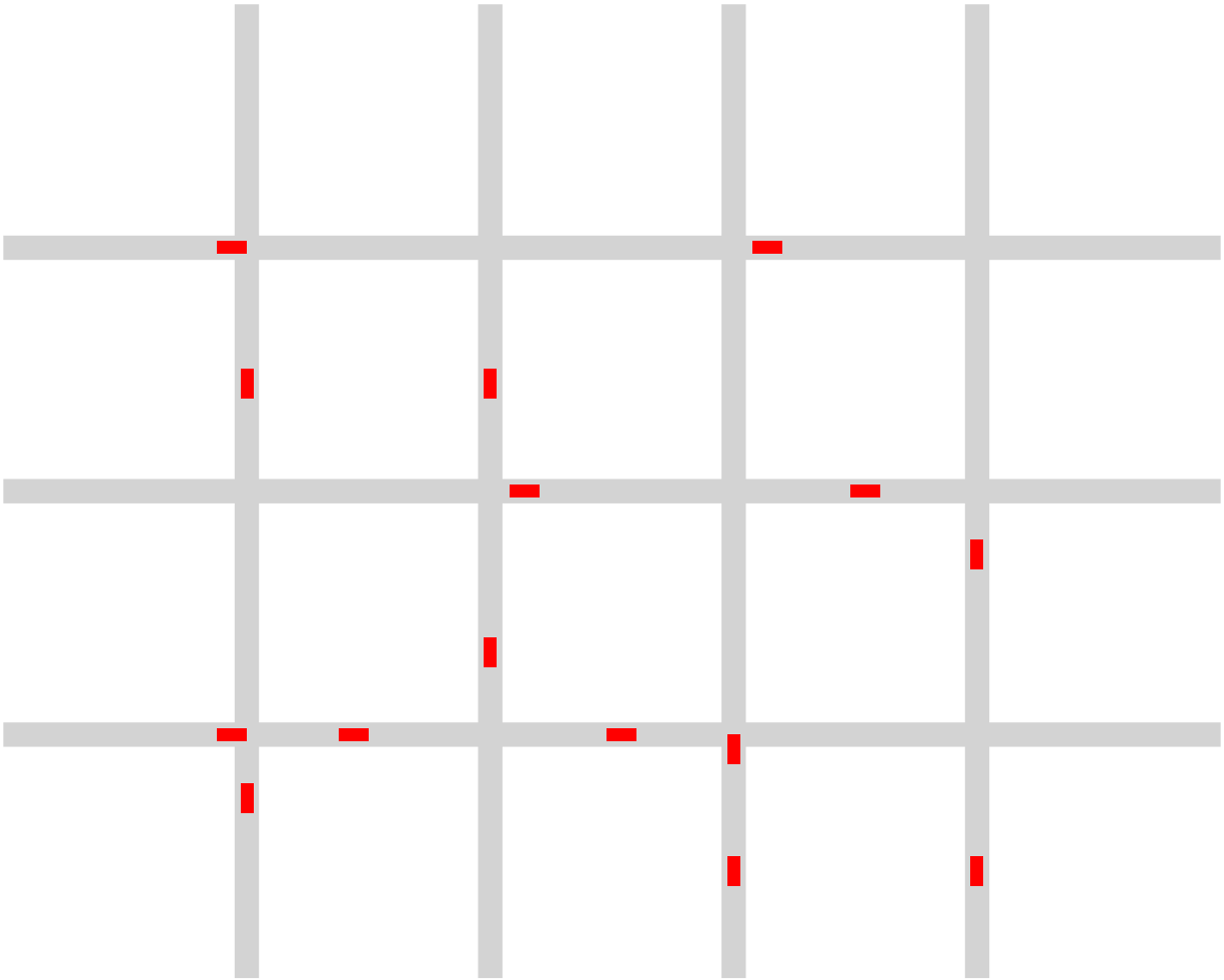

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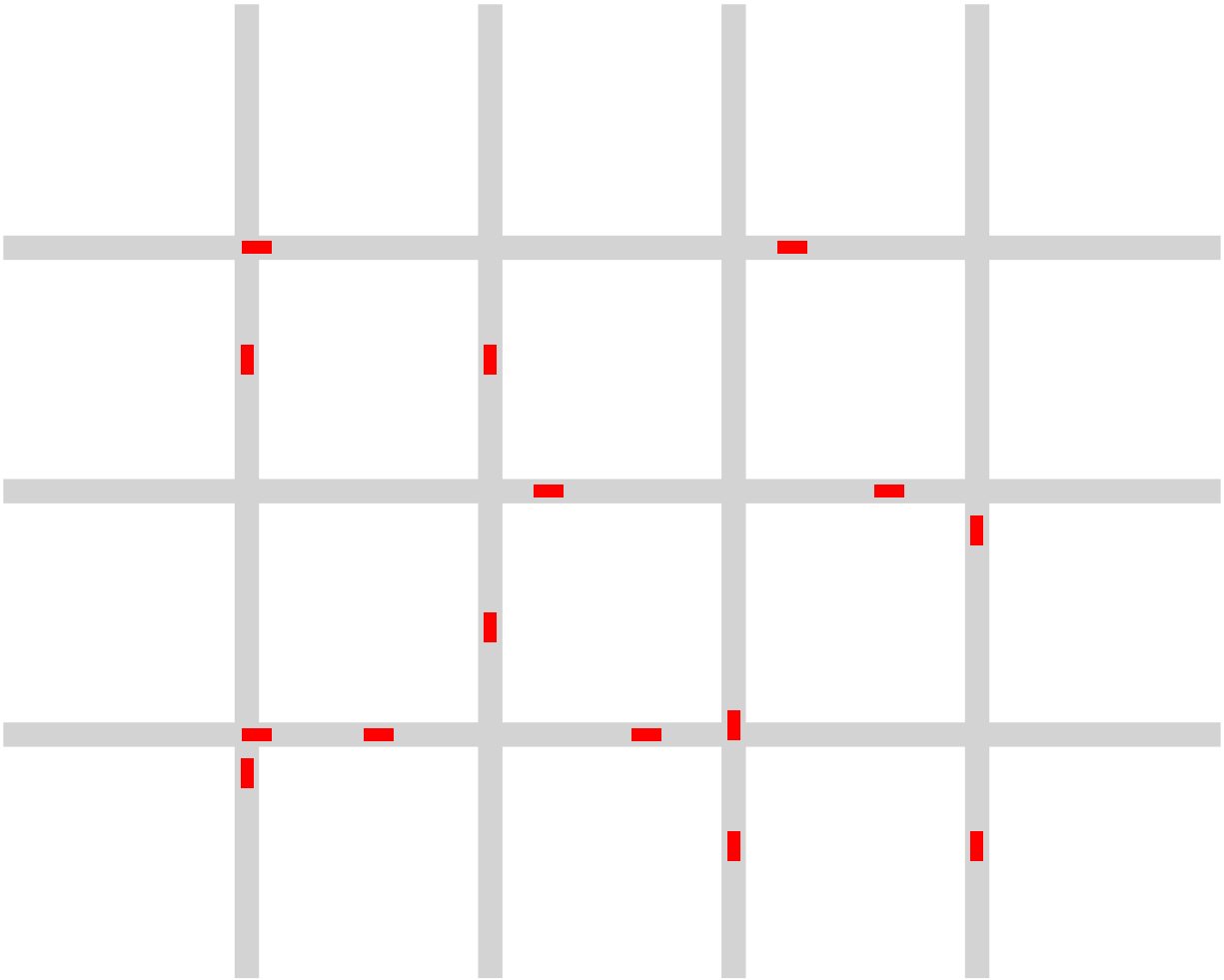
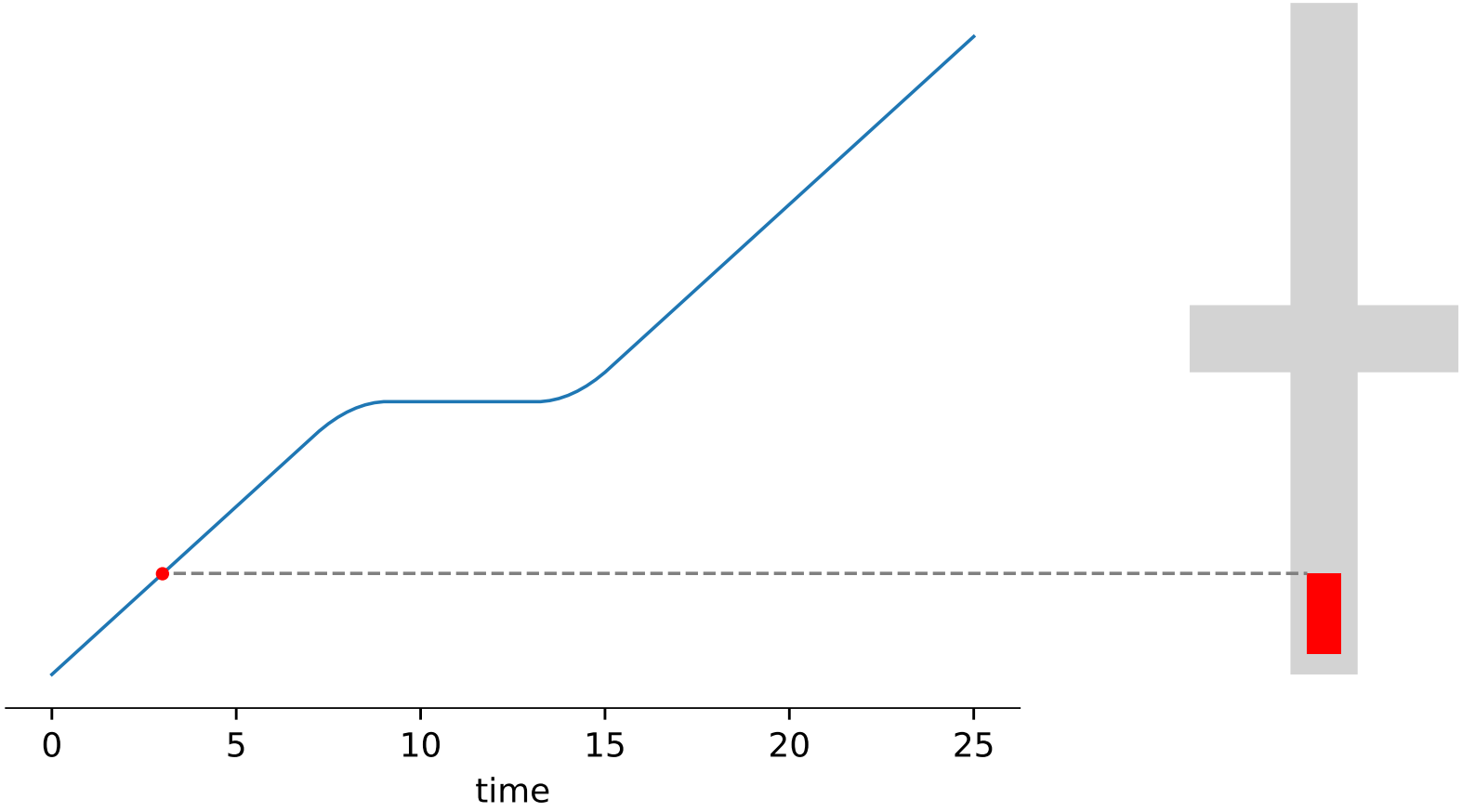


Outline

1. Crossing Time Scheduling
2. Cutting Planes in Single Intersection
3. Constructive Heuristics

Crossing Time Scheduling

Crossing Time



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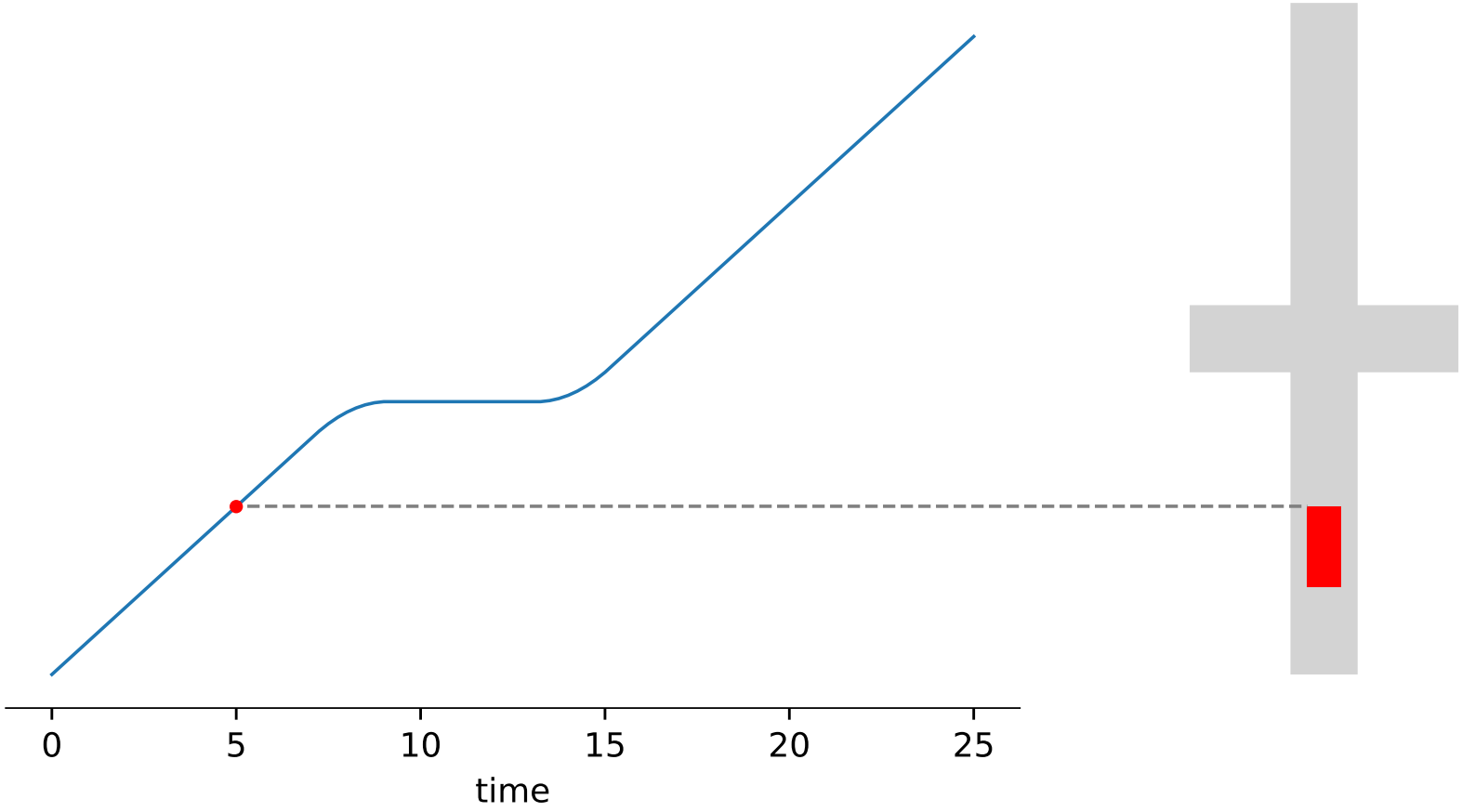

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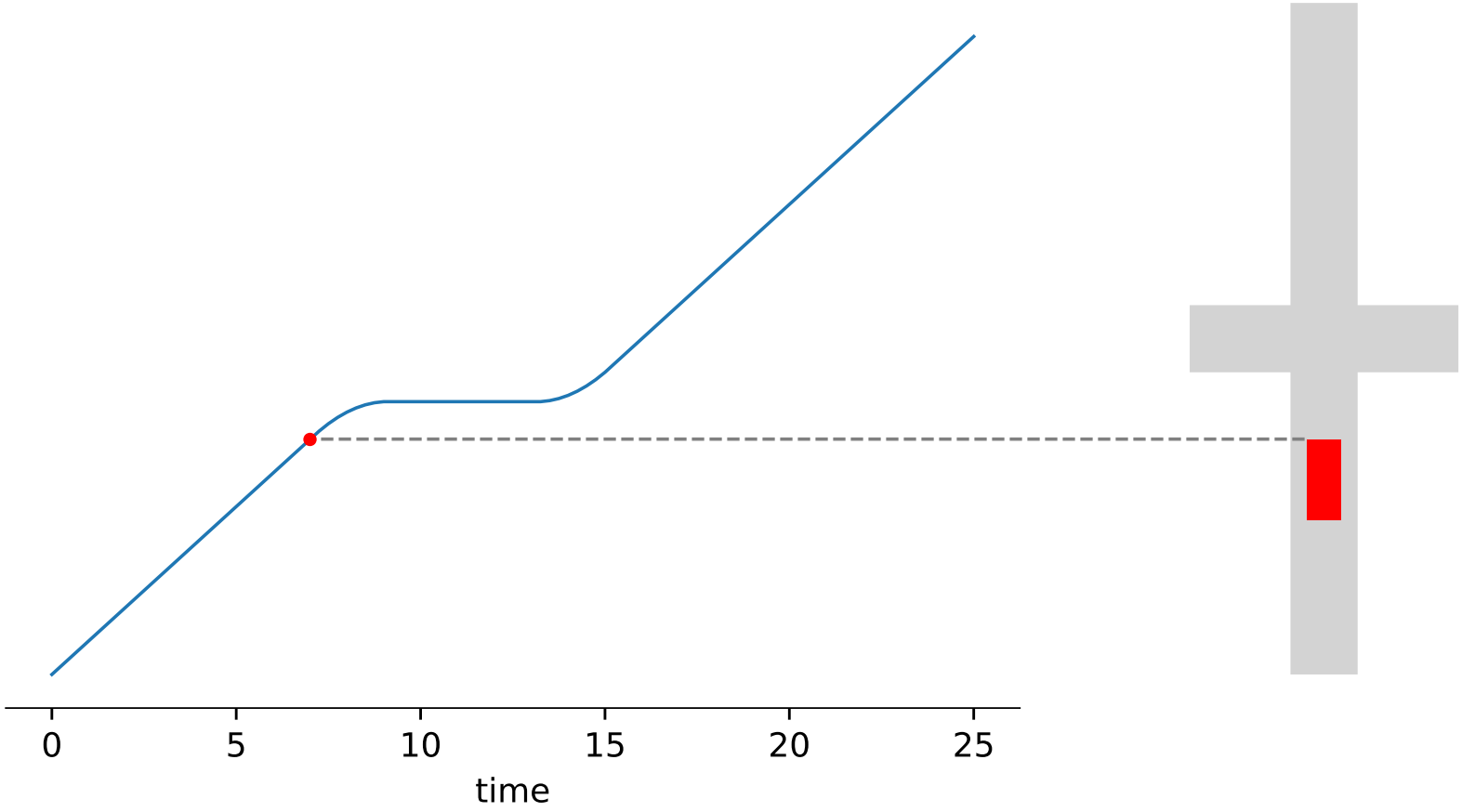

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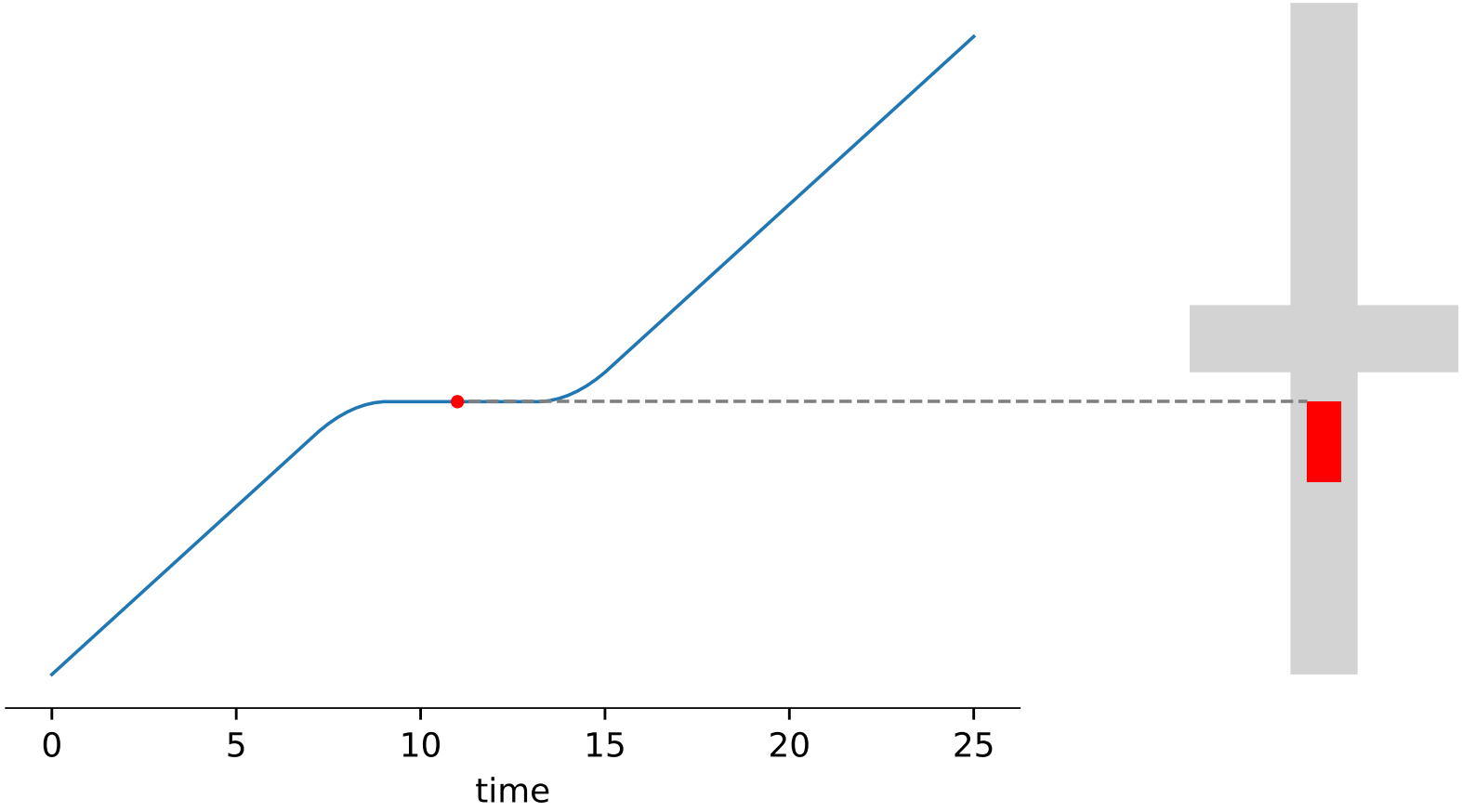

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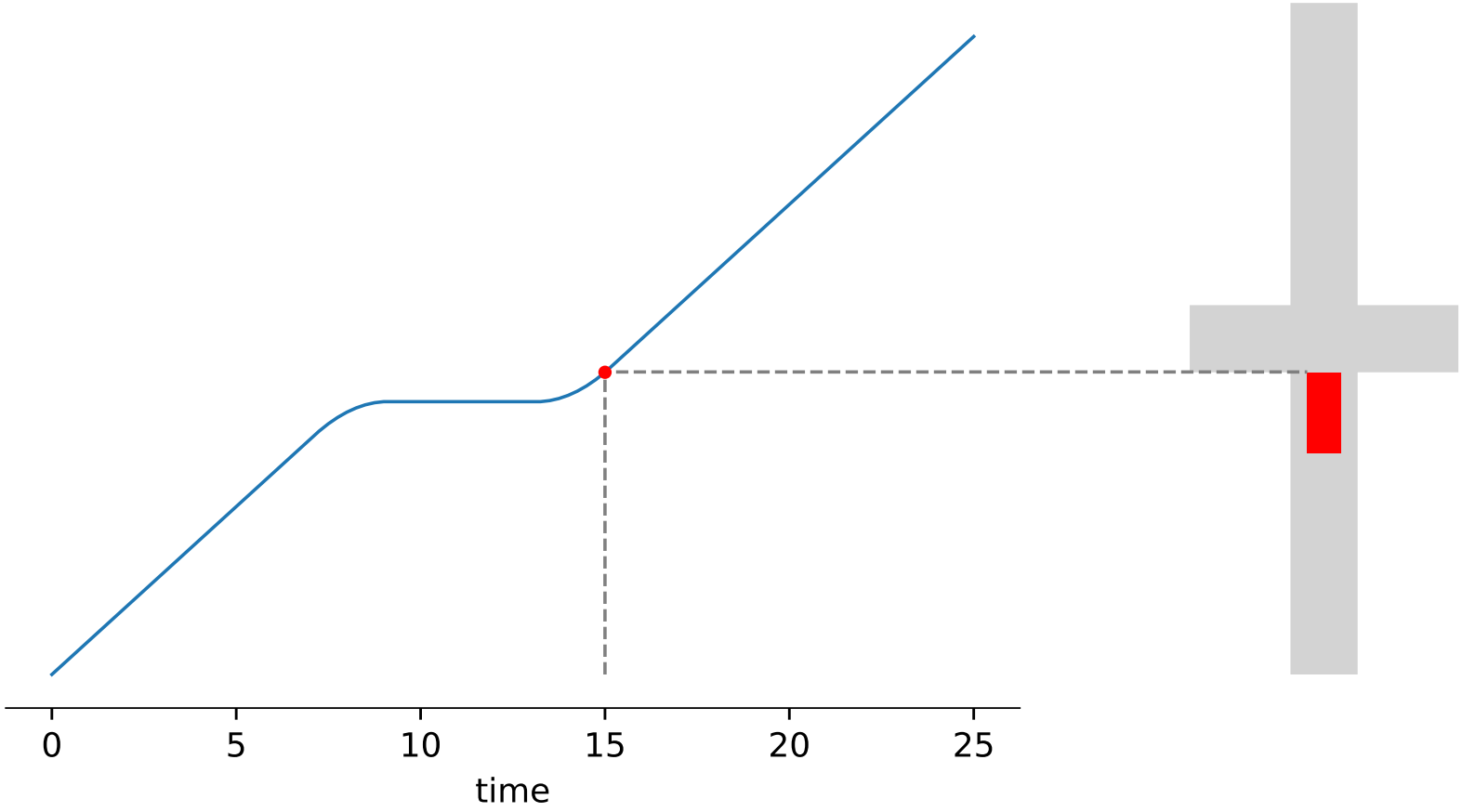

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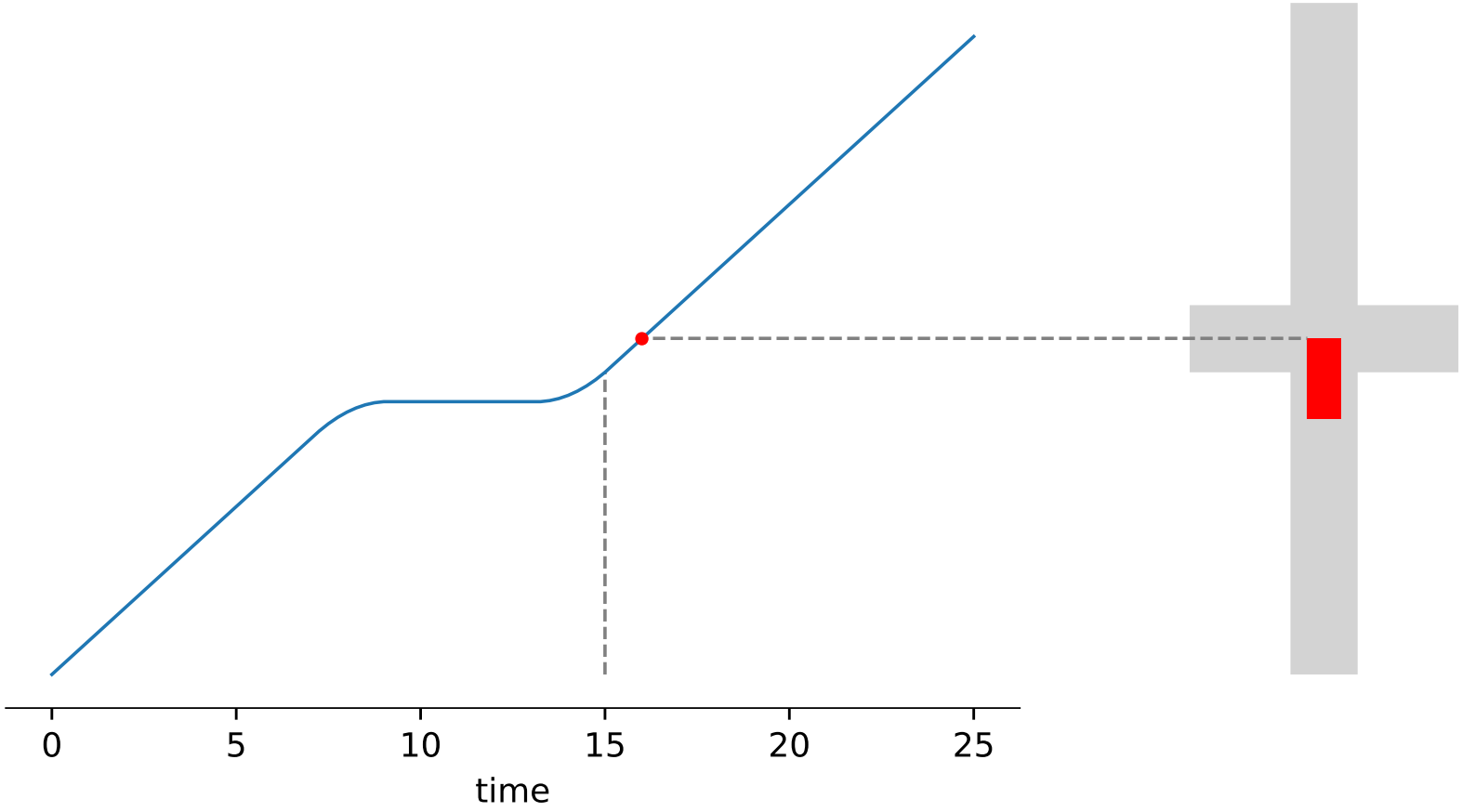

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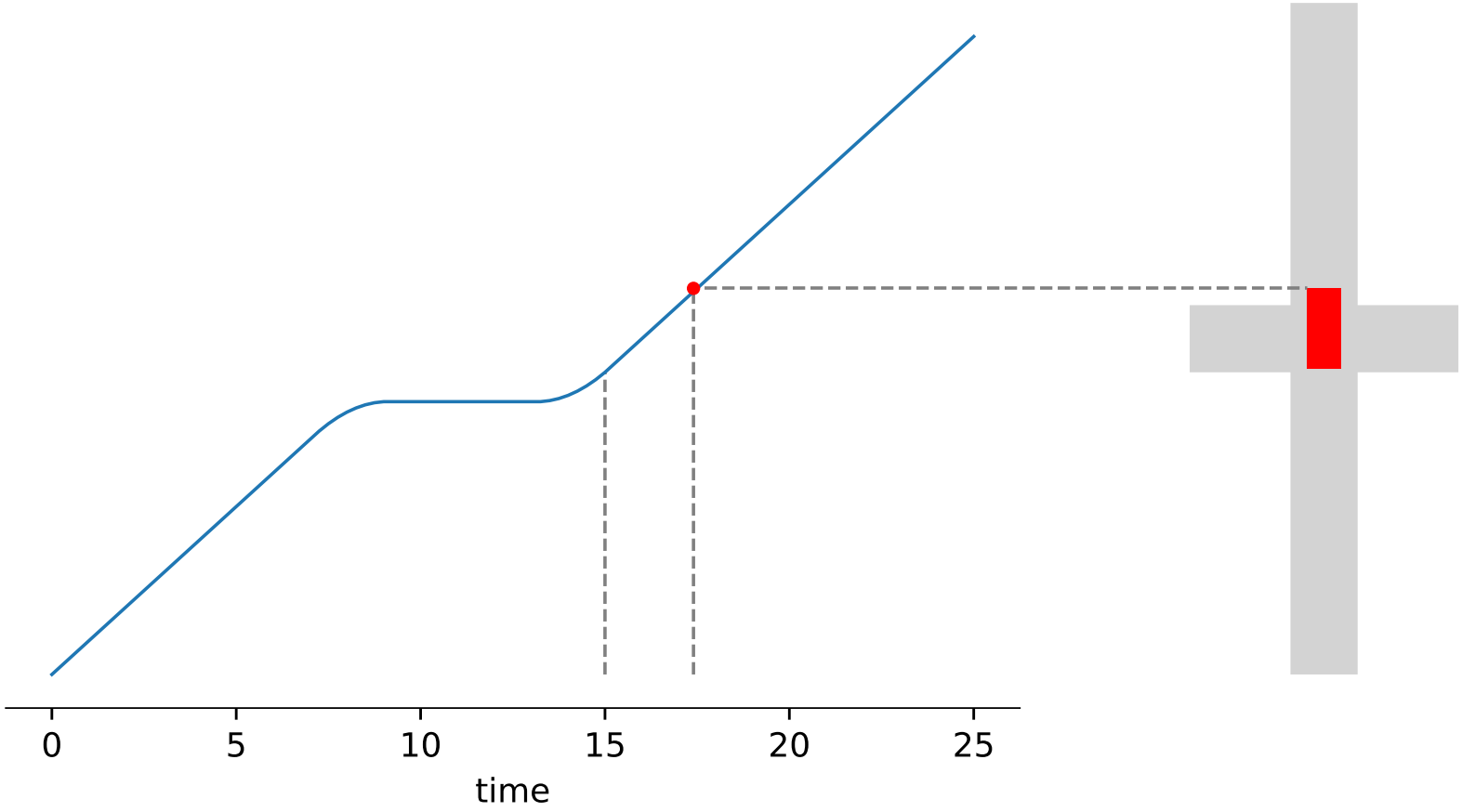

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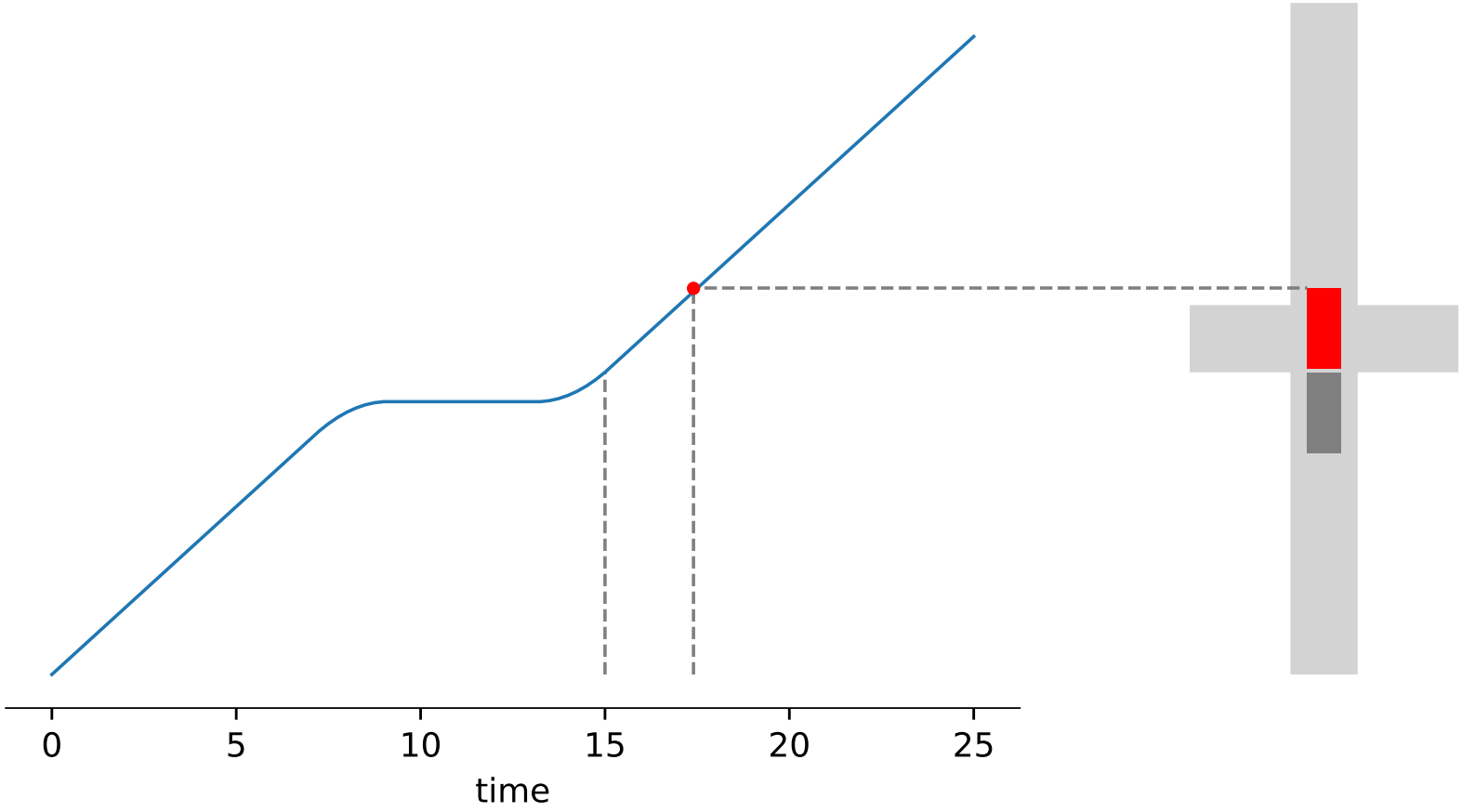

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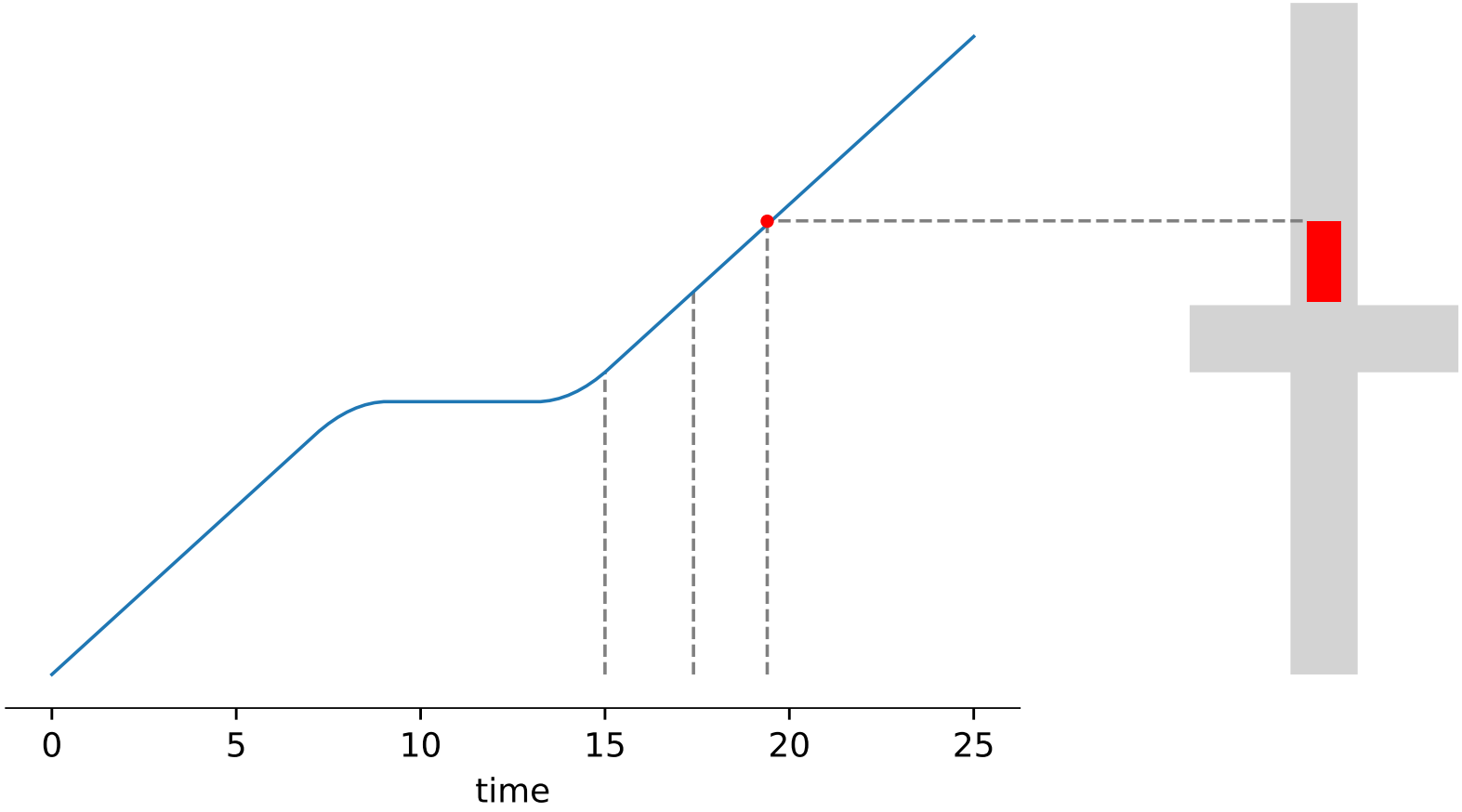

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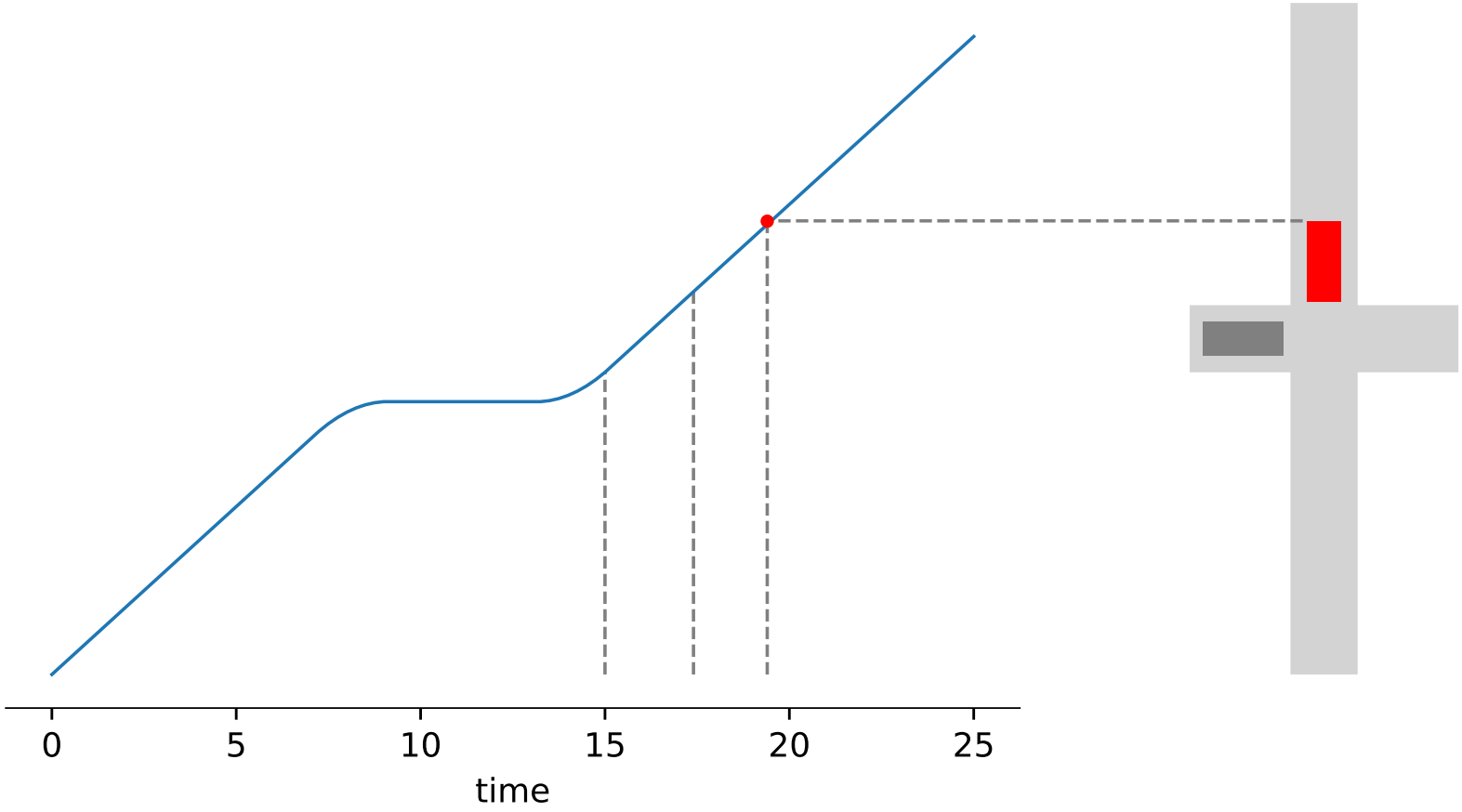

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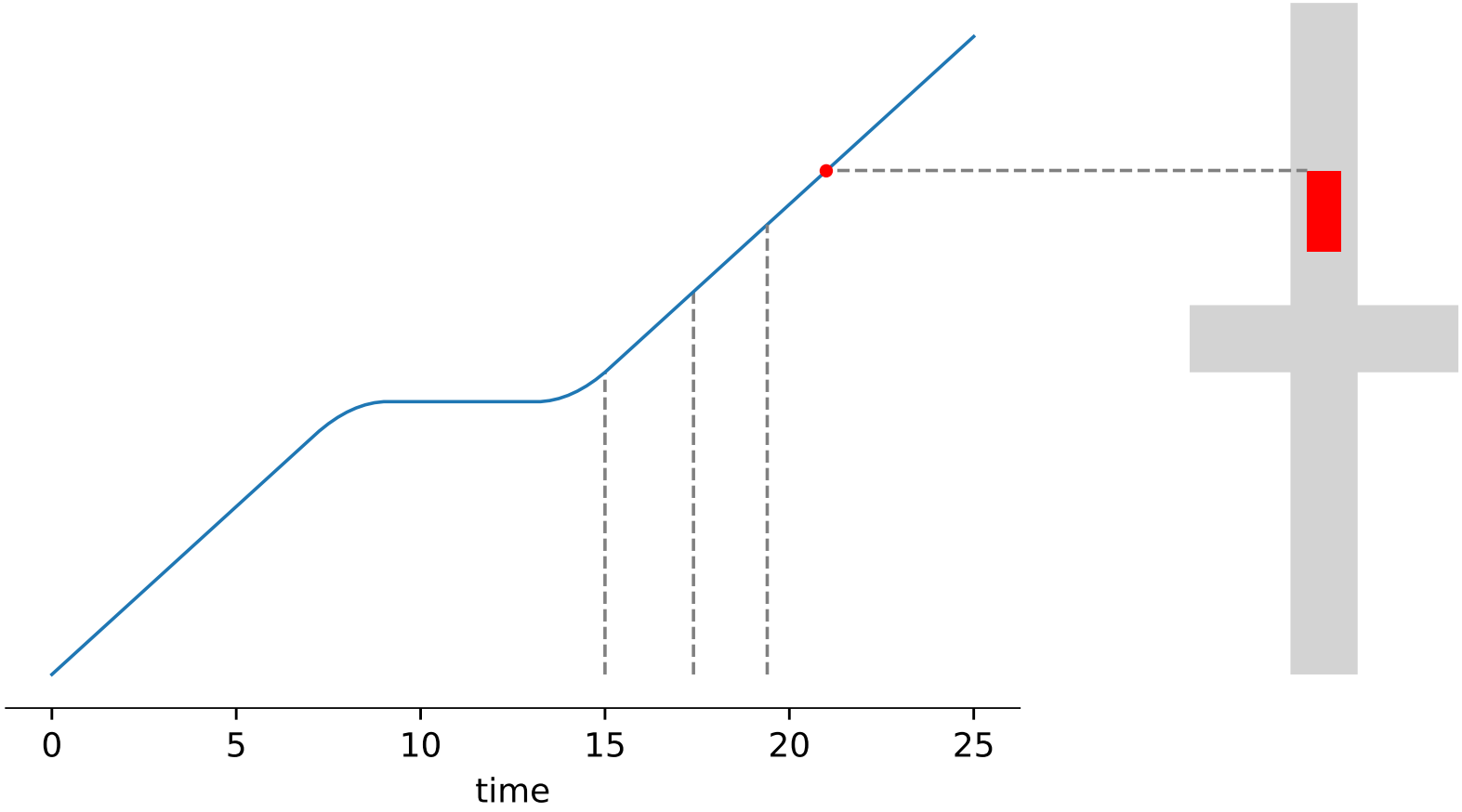

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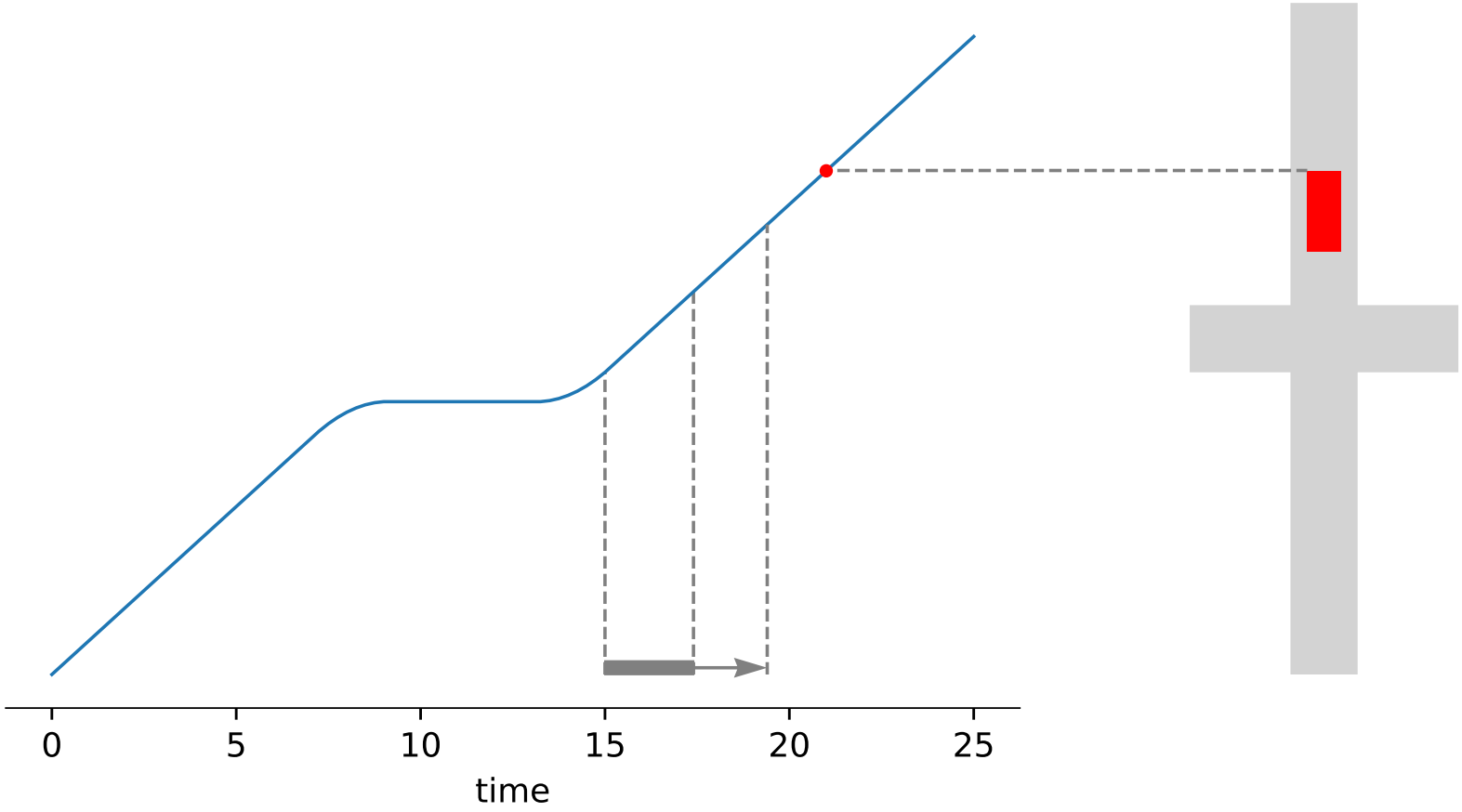
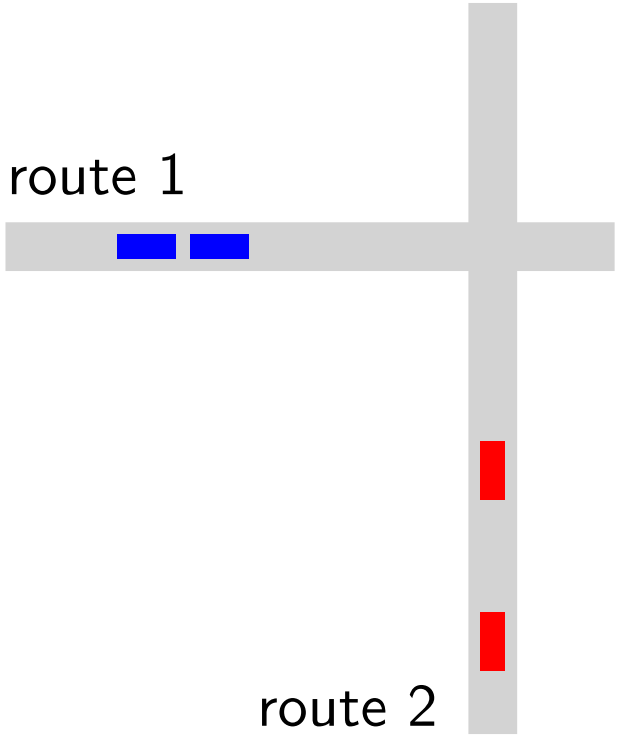


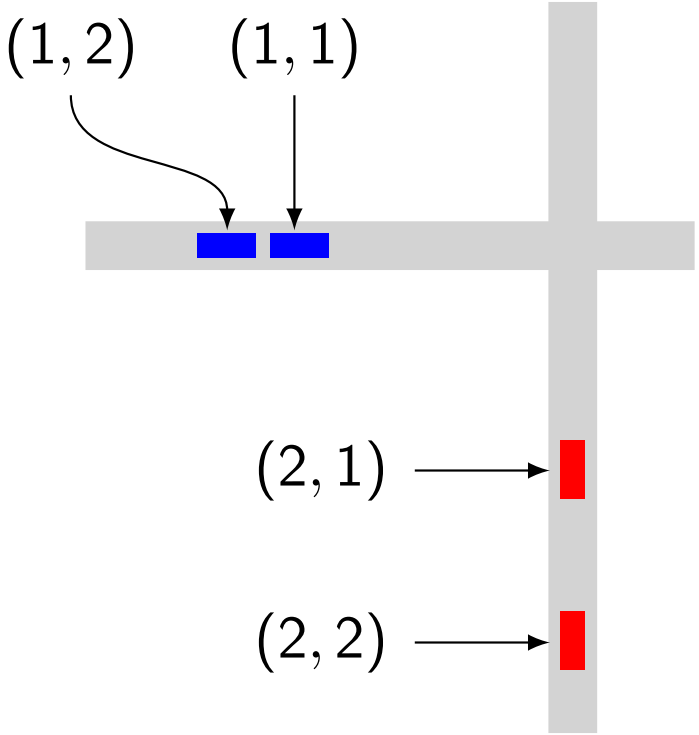
Crossing Time Scheduling

- Single intersection with two routes



Crossing Time Scheduling

- Vehicle $i = (r, k)$ is the kth vehicle on route r



Crossing Time Scheduling

- Vehicle $i = (r, k)$ is the kth vehicle on route r
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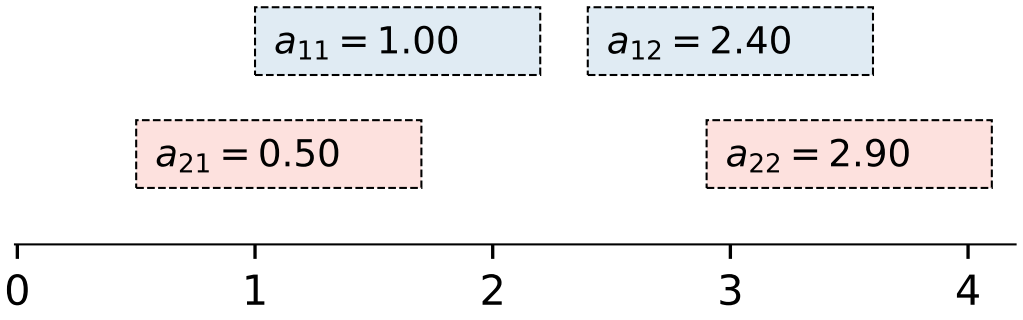

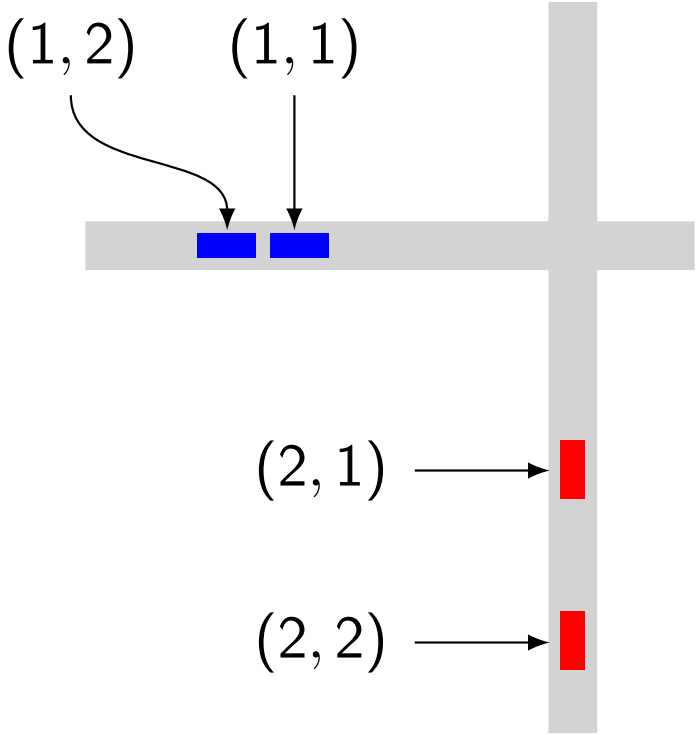

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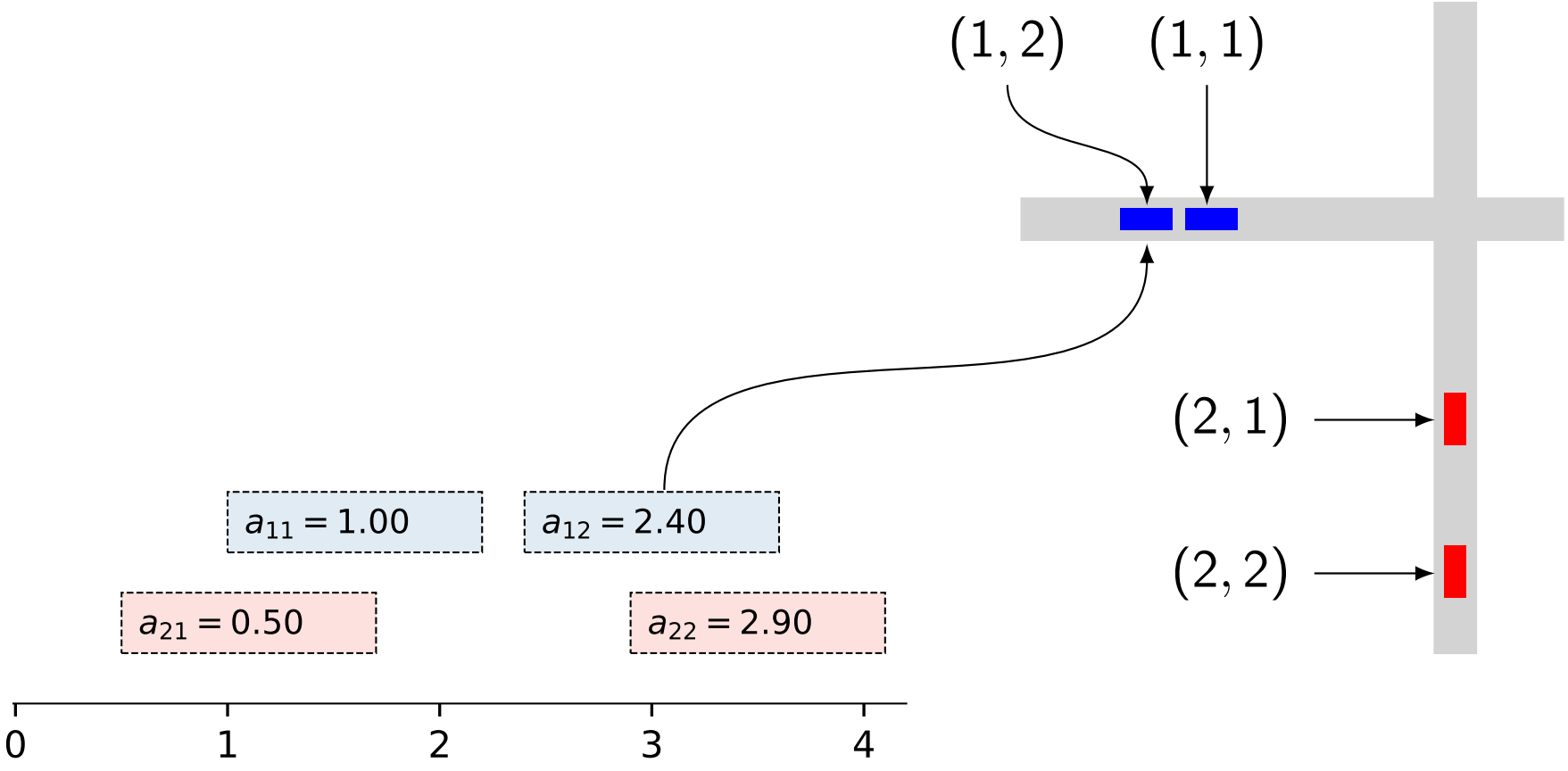

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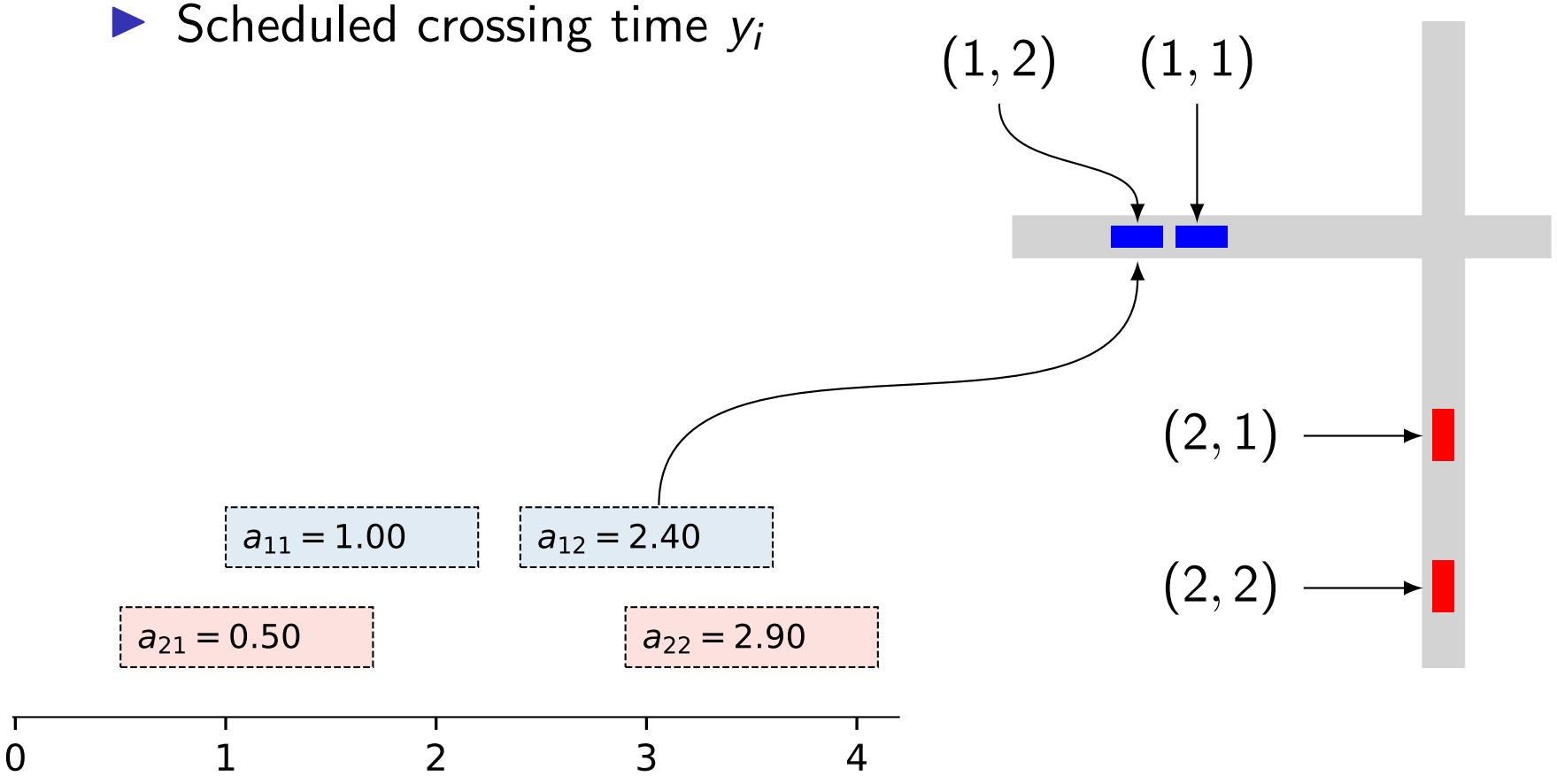

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- Vehicle $i = (r, k)$ is the kth vehicle on route r
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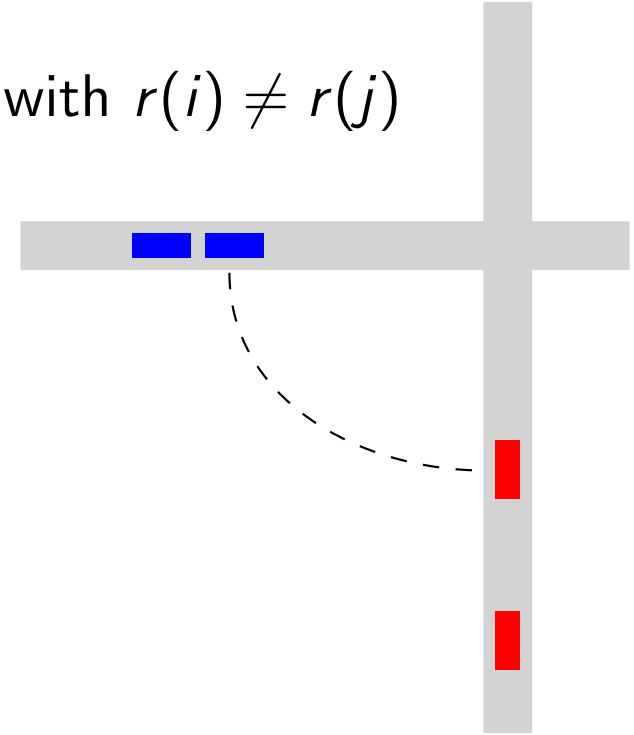

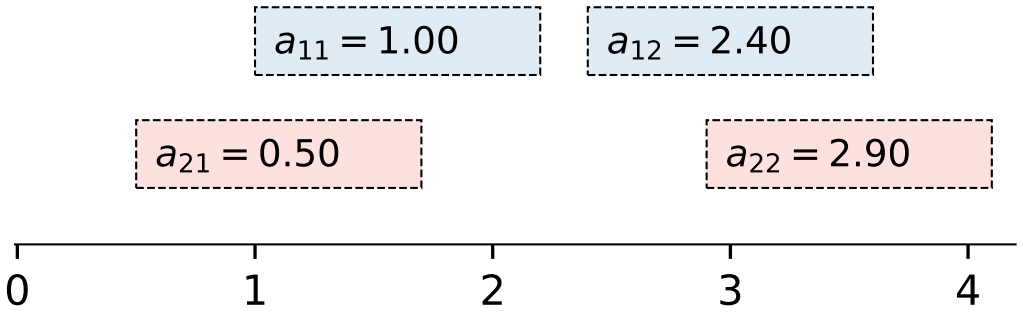

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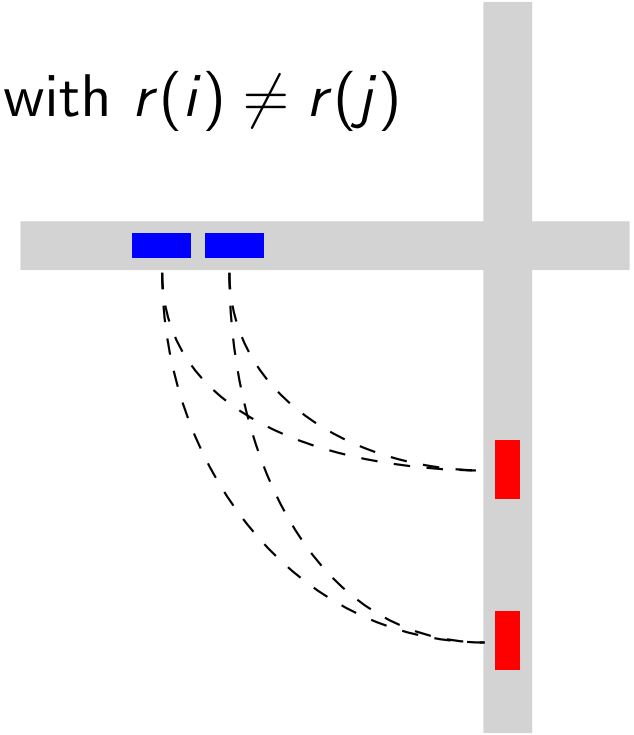

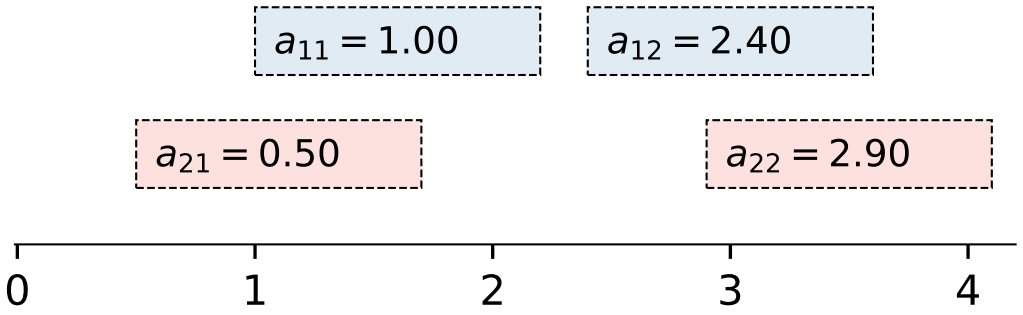

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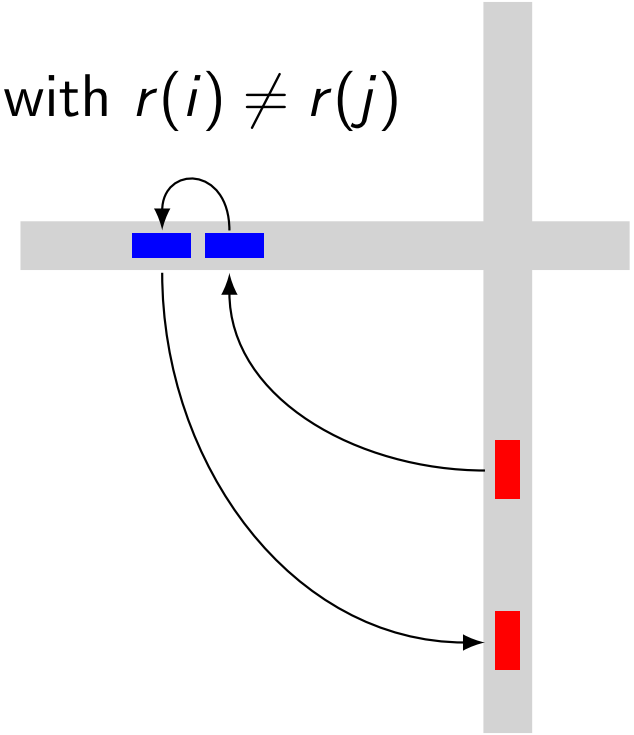

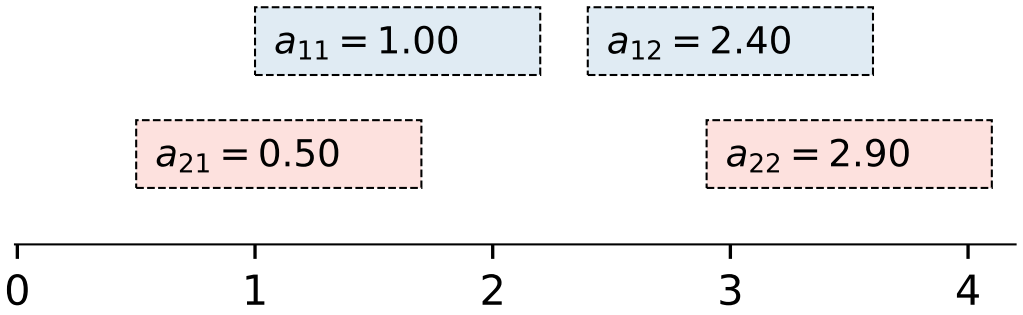

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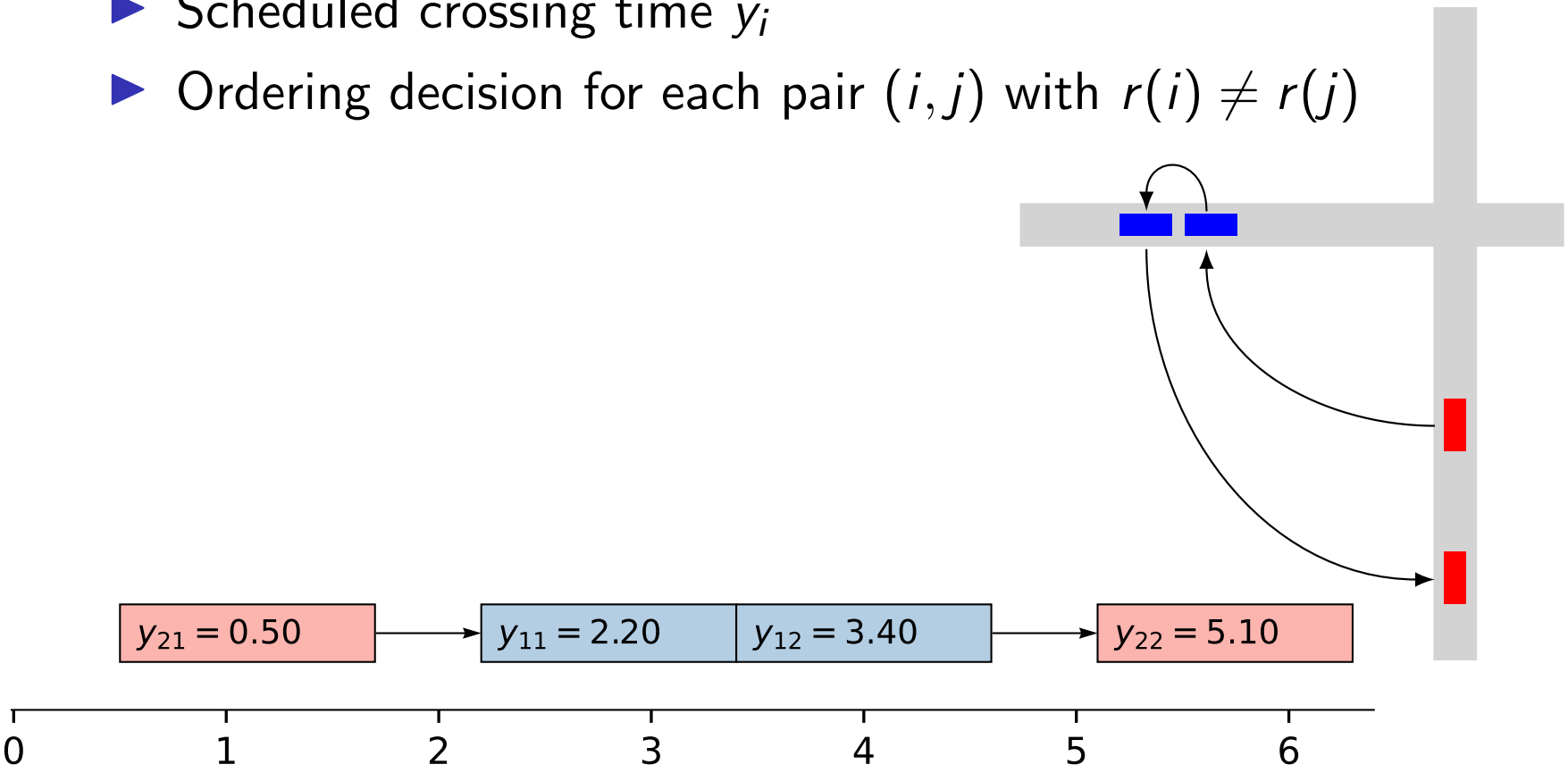

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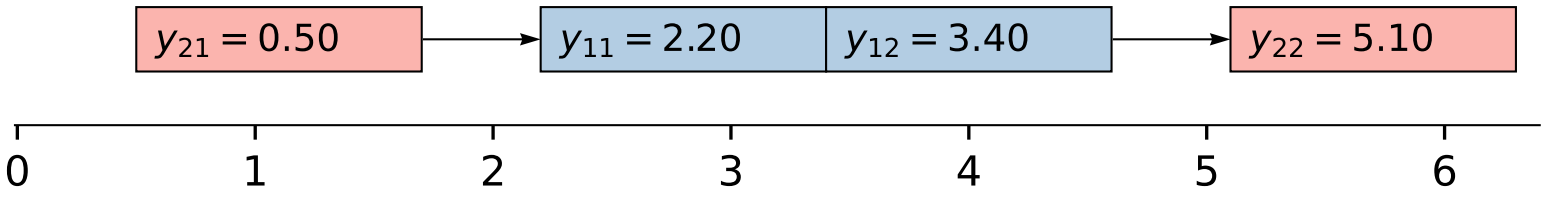

- Try to minimize total delay

$$\min_{y} \sum_{i} y_i - a_i$$

s.t. time slots are disjoint

Crossing Time Scheduling

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$$\min_{y} \sum_{i} y_i - a_i$$

$$\text{s.t. time slots are disjoint}$$

- Ordering decisions $\rightarrow$ Binary variables
- Mixed-Integer Linear Program (MILP)

Crossing Time Scheduling

- How fast can we solve this?

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- A couple of options
  - Reformulation
  - Primal heuristics
  - Cutting planes
  - Better branching decisions

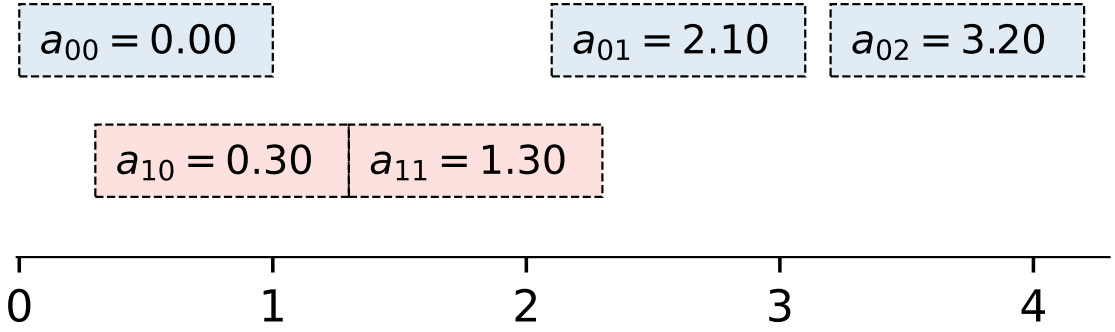
Crossing Time Scheduling

- How fast can we solve this?
- Vanilla integer programming does not scale so well

- A couple of options
  - Reformulation
  - Primal heuristics
  - Cutting planes $\longleftarrow$ There are some for a single intersection!
  - Better branching decisions

Cutting Planes in Single Intersection

Platoon Preservation



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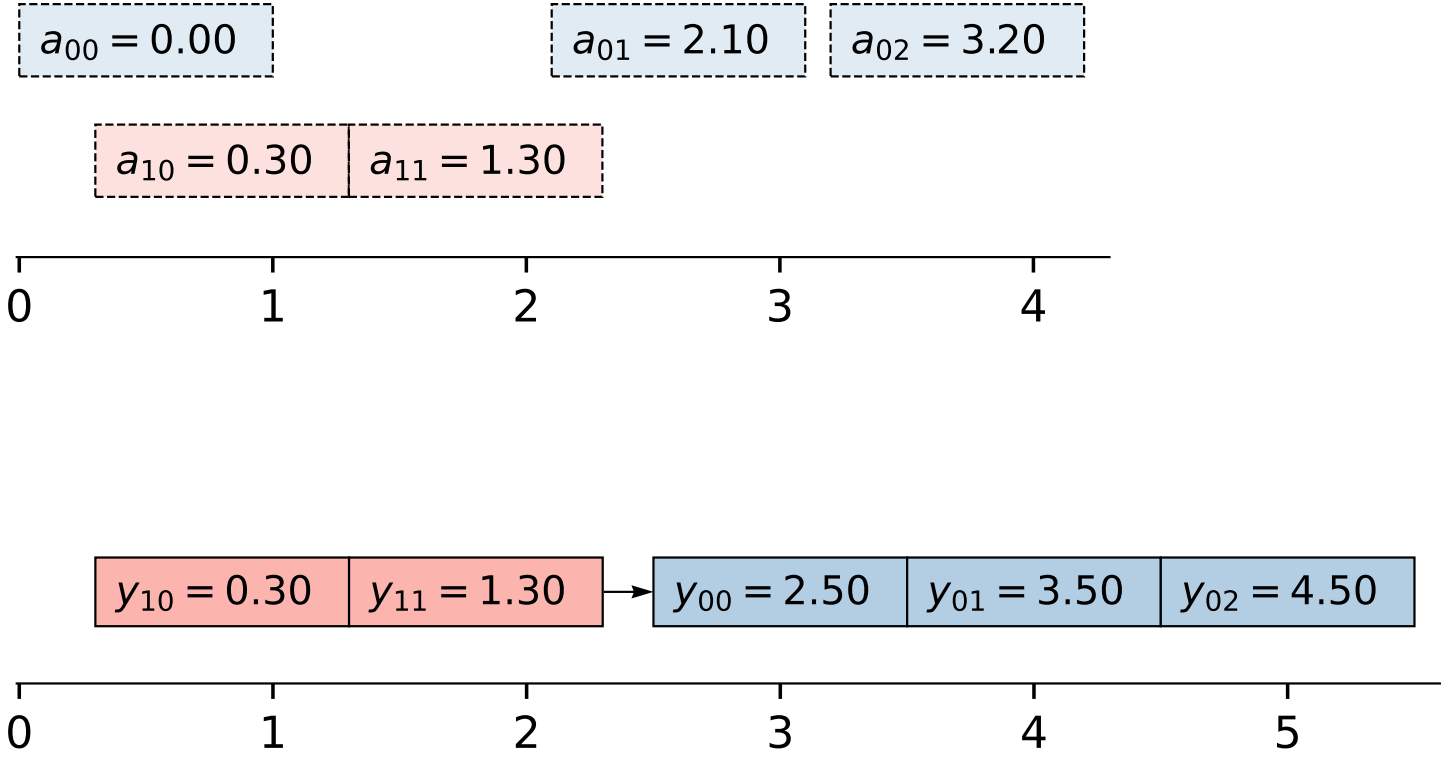

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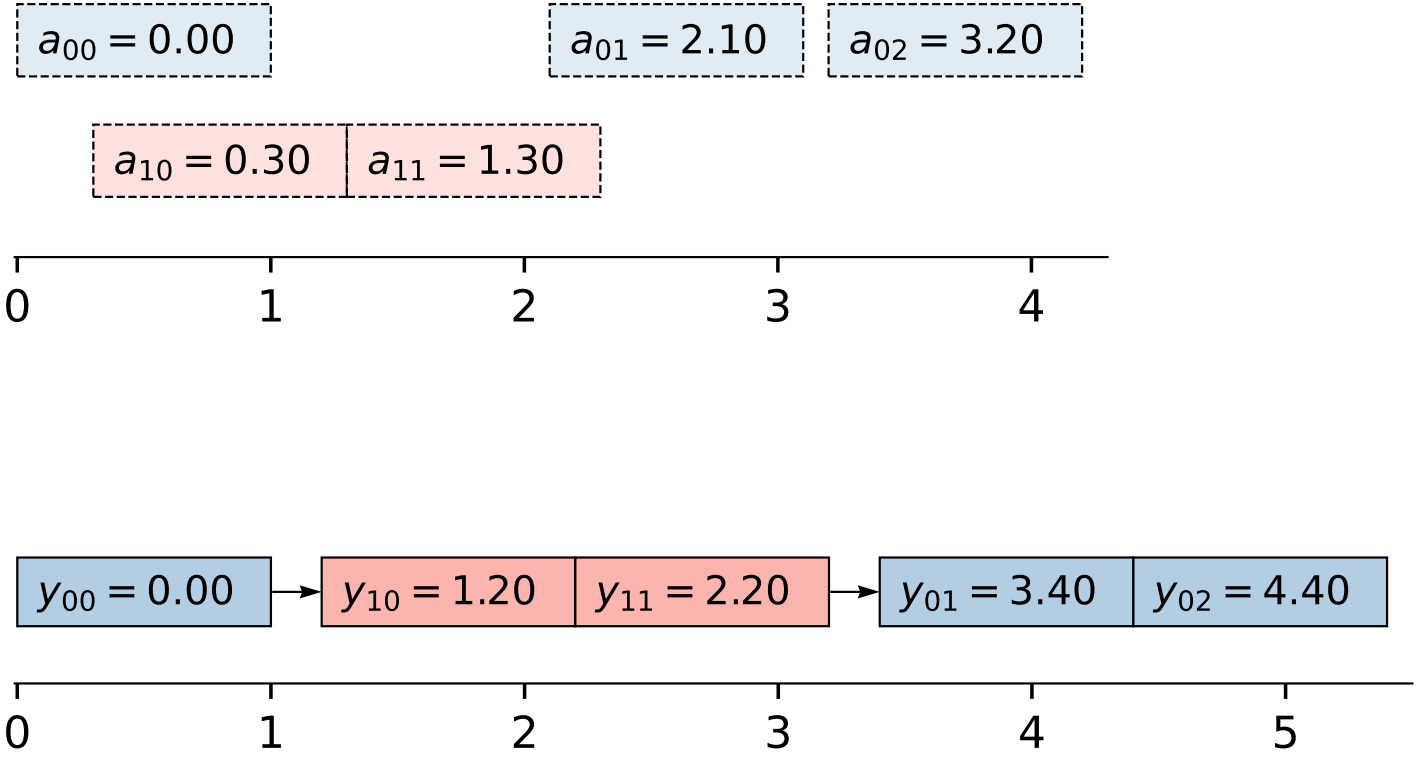

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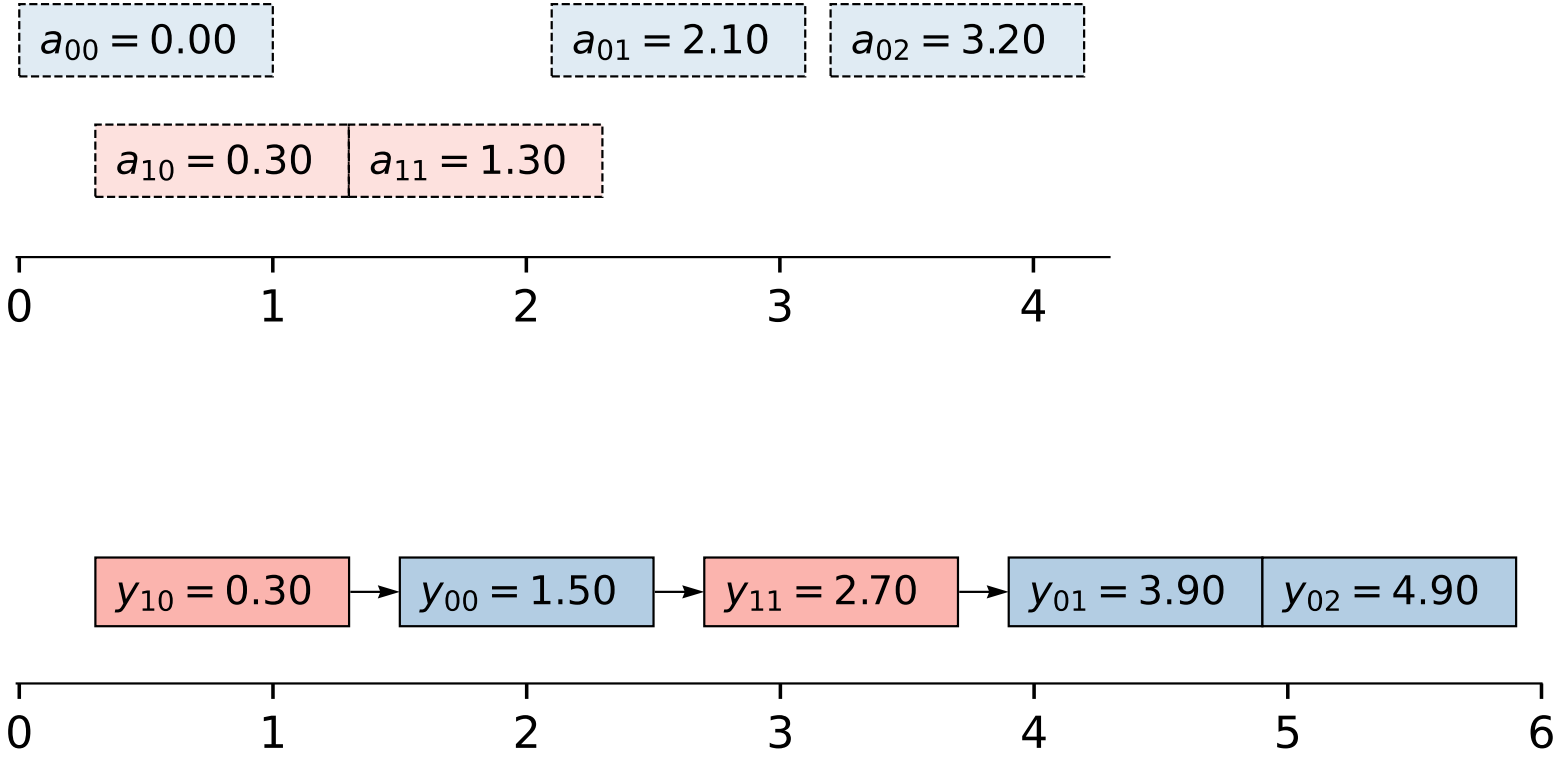

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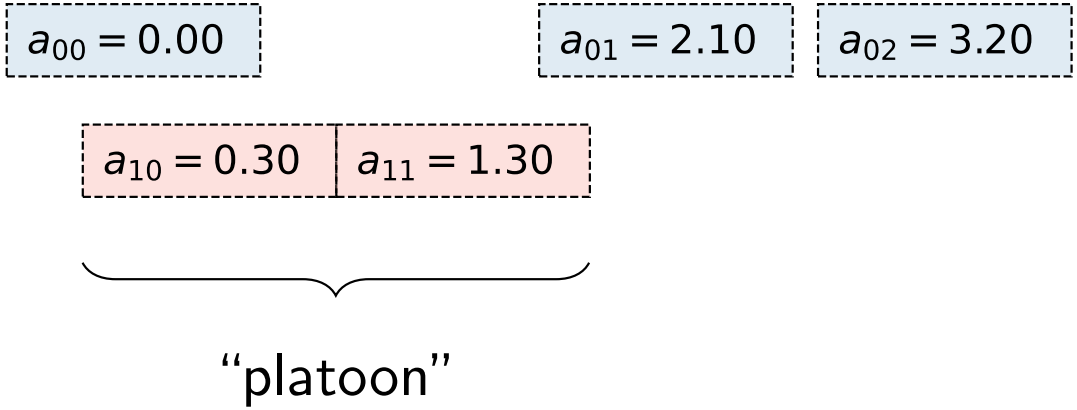
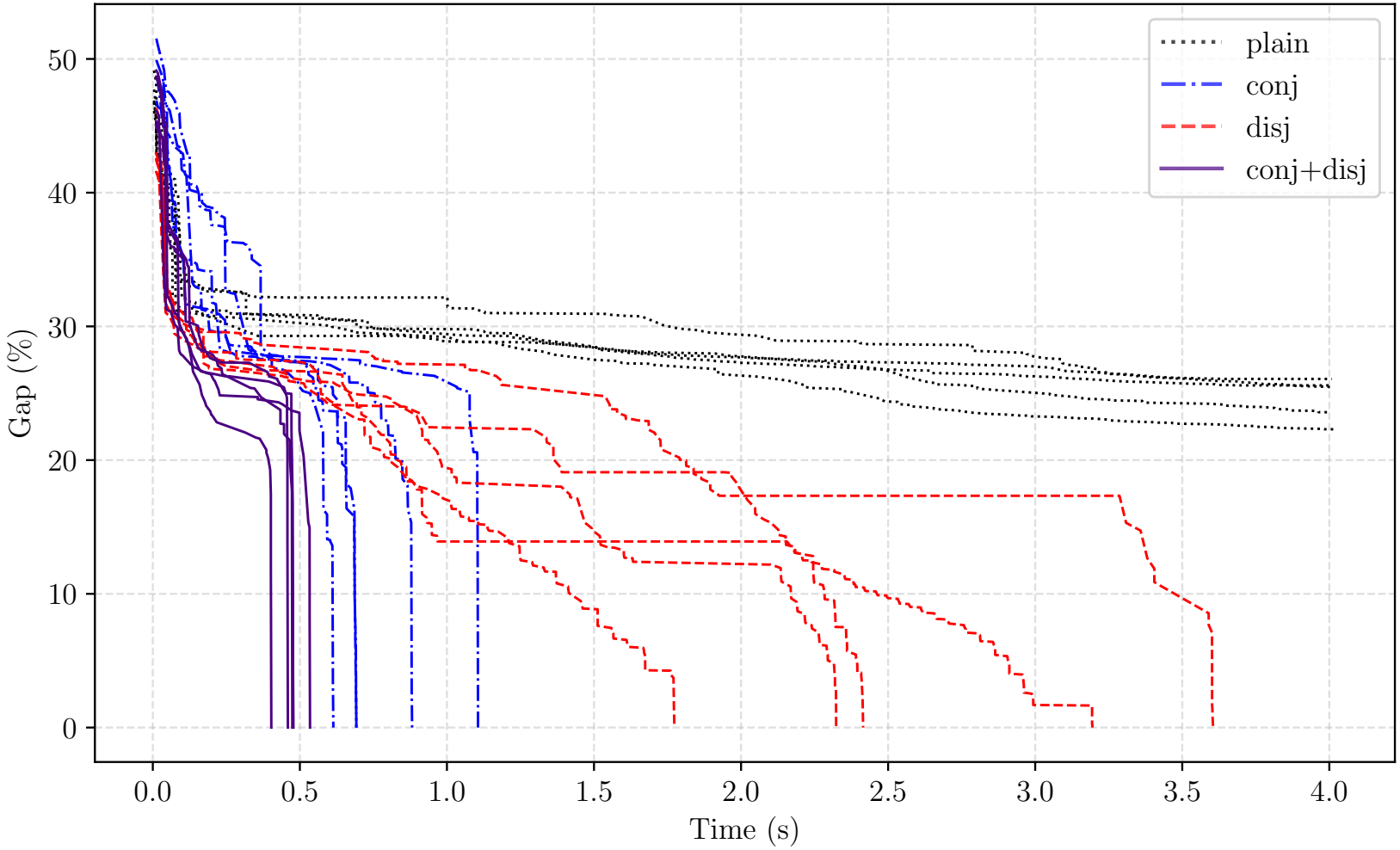


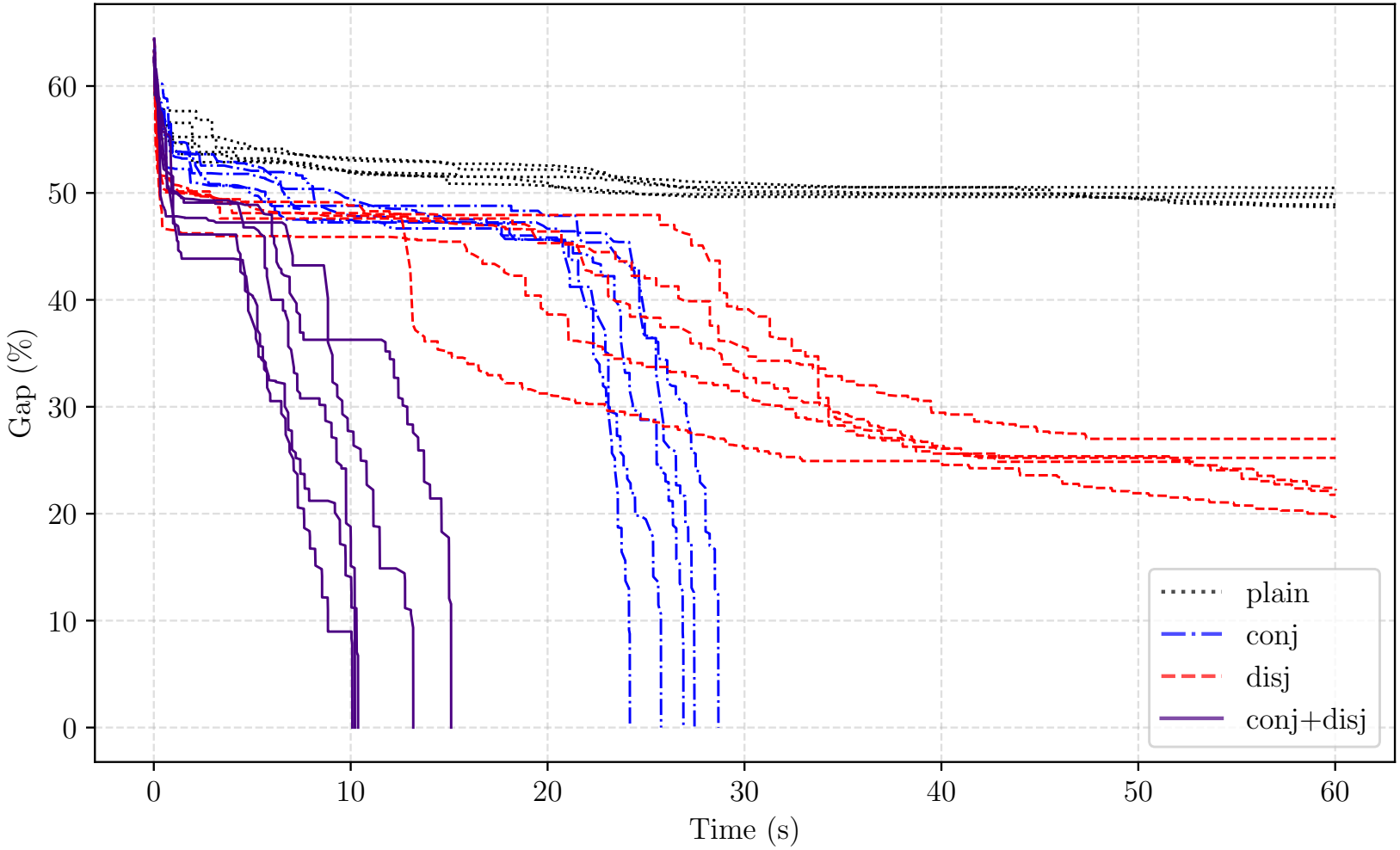
Cutting Planes in Single Intersection

- Two routes, 20 vehicles per route, 40 in total



Cutting Planes in Single Intersection

- Three routes, 20 vehicles per route, 60 in total



Constructive Heuristics

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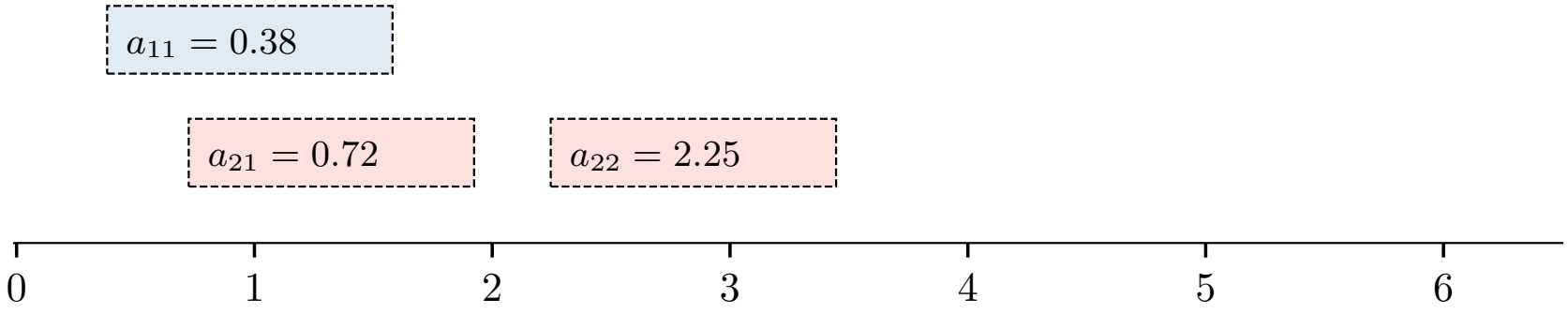
- Construct the schedule step by step

Constructive Heuristics



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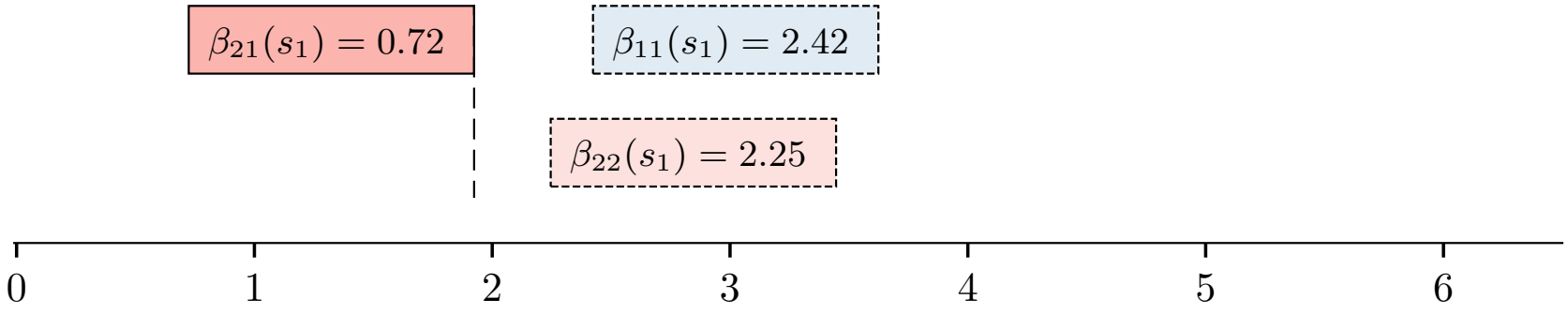

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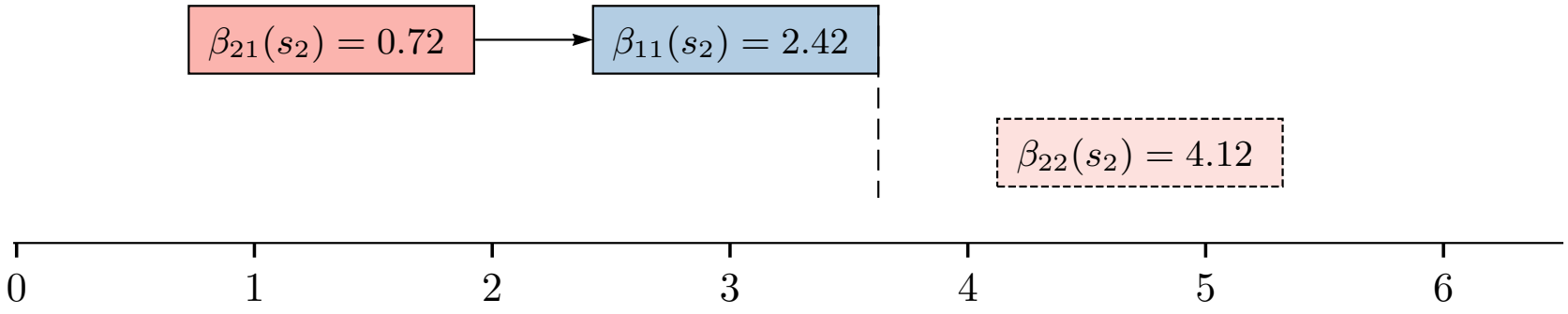

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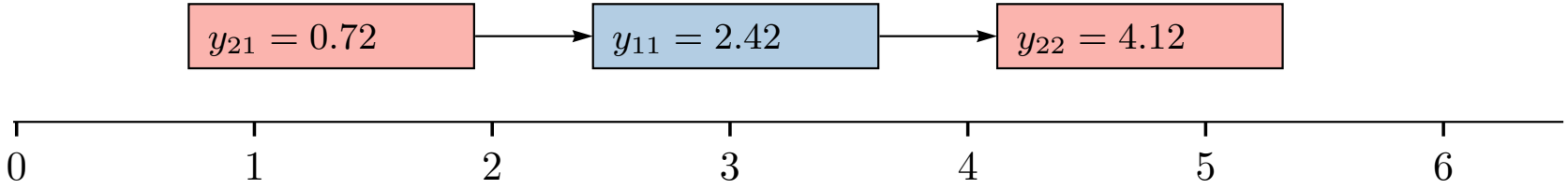
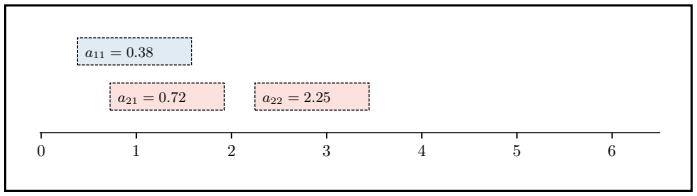


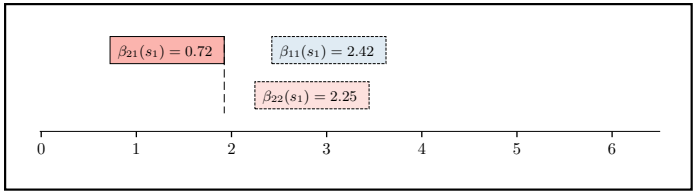
Formulate as MDP

- Action = route index
- Reward = change in delay

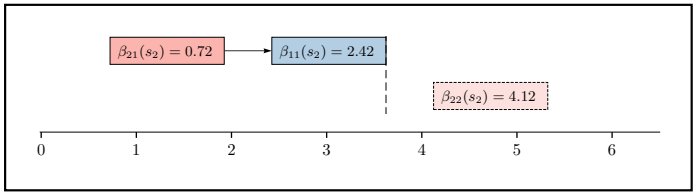
$s_1 =$



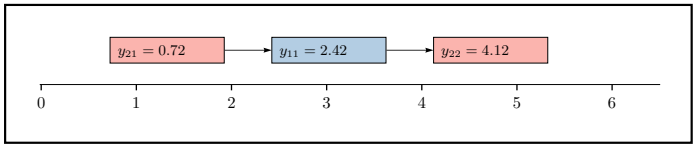
$s_2 =$



$s_3 =$



$s_4 =$



Formulate as MDP

- Action = route index
- Reward = change in delay

$\Longrightarrow$ Final reward = total delay

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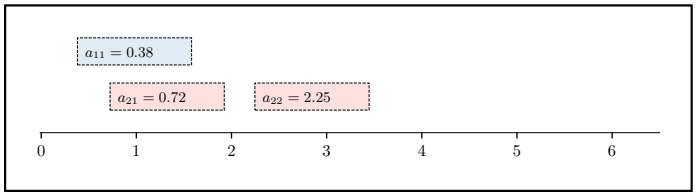

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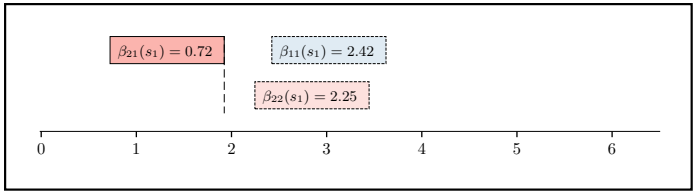
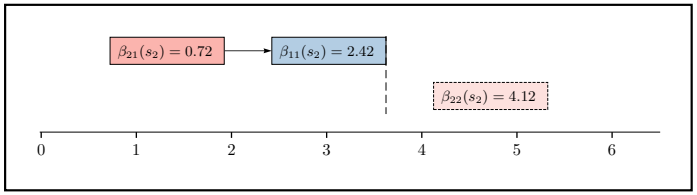


$s_3 =$



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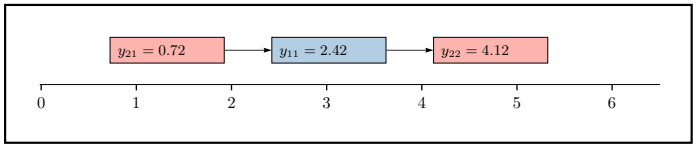
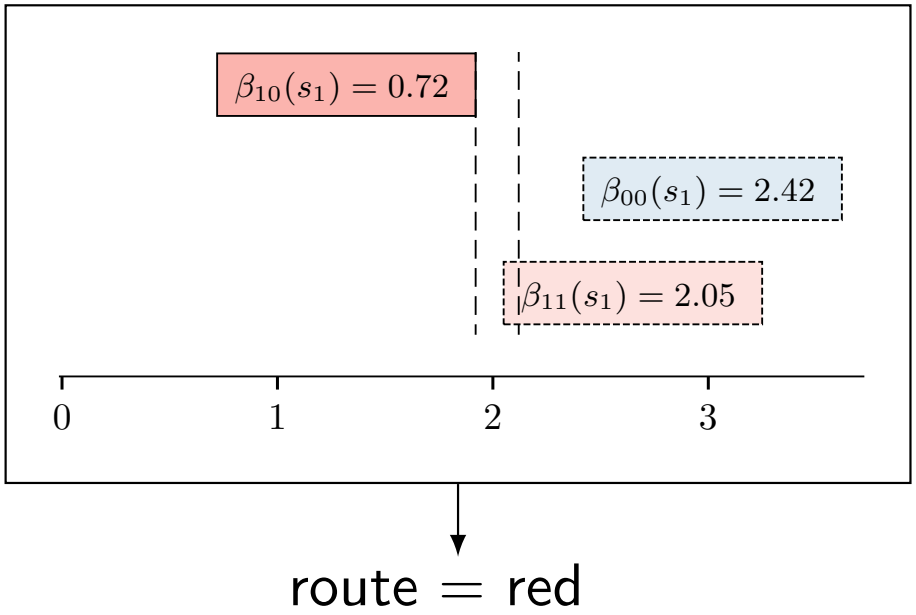
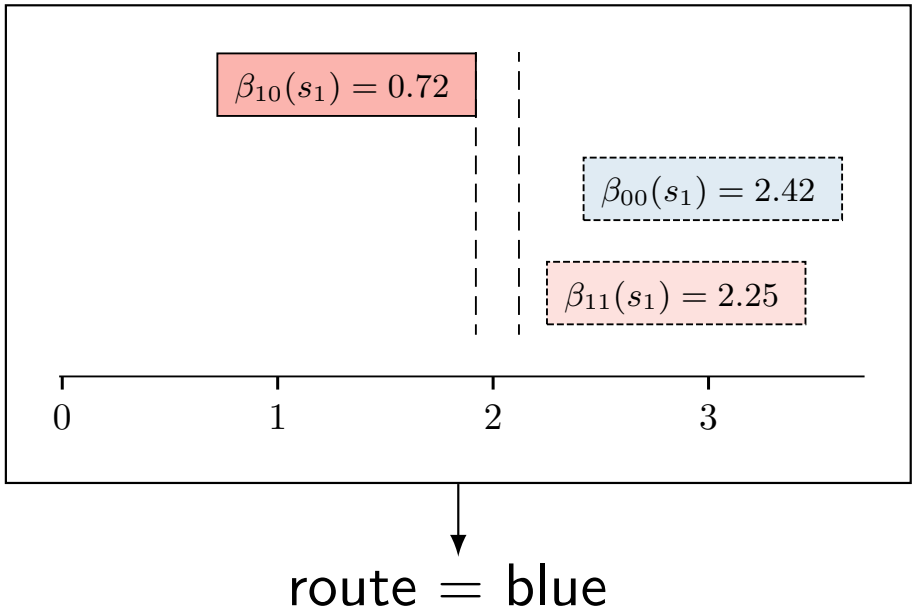
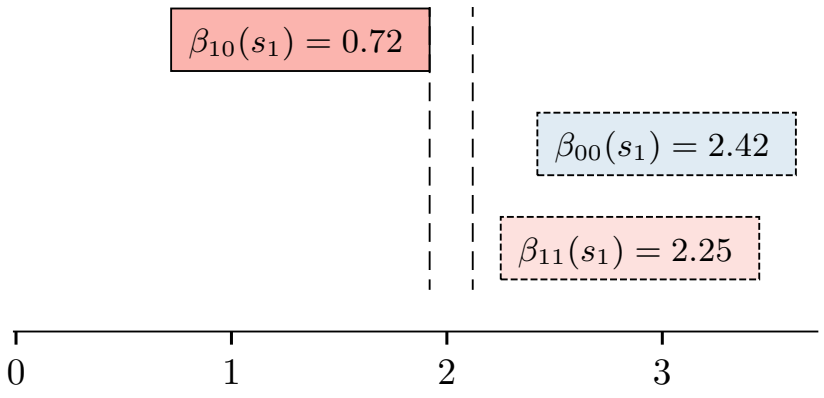


Threshold policy

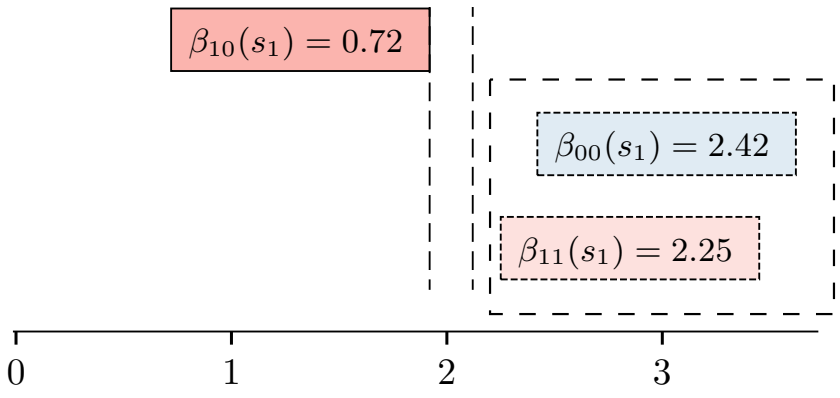


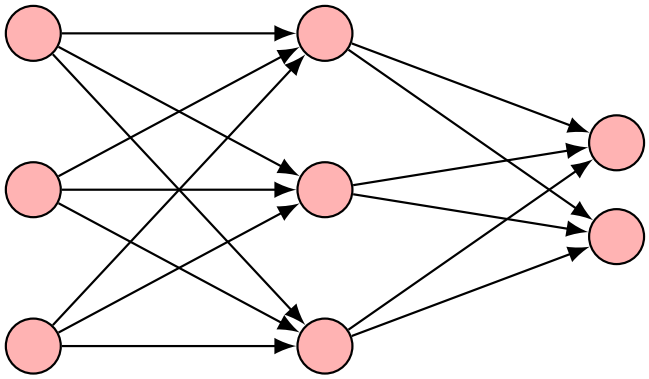


Neural parameterized policy



Neural parameterized policy





- Imitation learning
- Policy-gradient optimization

		MILP		Threshold			Imitation			PPO		
n_r	R	opt	T	obj	gap	T_{train}	obj	gap	T_{train}	obj	gap	T_{train}
20	2	13.54	(3s)	13.64	**0.75**%	(6s)	13.74	1.54%	(12s)	14.00	3.41%	(2m)
40	2	26.35	(17s)	26.76	**1.53**%	(12s)	26.97	2.35%	(19s)	26.96	2.29%	(2m)
60	2	41.89	(27s)	42.63	1.77%	(18s)	42.57	**1.62**%	(23s)	43.54	3.94%	(2m)
20	3	60.47	(1m)	65.84	**8.89**%	(11s)	67.59	11.79%	(53s)	69.91	15.62%	(2m)
40	3	121.91	(15m)	132.56	8.73%	(21s)	132.14	**8.39**%	(3m)	151.01	23.87%	(2m)
60	3	187.47	(38m)	202.12	**7.81**%	(32s)	202.73	8.14%	(6m)	233.93	24.78%	(2m)

Conclusion

- Single intersection $\rightarrow$ cutting planes
- Constructive scheduling $\rightarrow$ threshold heuristic

- Further questions
  - Cutting planes in networks?
  - Constructive scheduling in networks?
  - Remove full speed assumption